

Topics on Artificial Intelligence & Machine Learning

D. T. McGuiness, Ph.D

B.Sc Data Science I Lecture Book

SS.2025



Table of Contents

Part I Probability & Statistics

Chapter 1	Theory of Probability	3
1.1	Introduction	3
	Representing Data · Graphic Representation of Data · Experiments & Outcomes · Unions, Intersections, and Complements of Events	
1.2	Probability	10
1.3	Permutations & Combinations	16
	Permutations · Combinations · Factorial Function · Binomial Coefficients	
1.4	Random Variables and Probability Distributions	20
	Discrete Random Variables and Distributions · Continuous Random Variables and Distributions	
1.5	Mean and Variance of a Distribution	25
1.6	Binomial, Poisson, and Hyper-geometric Distributions	28
	Sampling with Replacement · Sampling without Replacement: Hyper-geometric Distribution	
1.7	Normal Distribution	32
	Distribution Function · Numeric Values · Normal Approximation of the Binomial Distribution	
1.8	Distribution of Several Random Variables	36
	Discrete Two-Dimensional Distribution · Continuous Two-Dimensional Distribution · Marginal Distributions of a Discrete Distribution · Marginal Distributions of a Continuous Distribution · Independence of Random Variables · Functions of Random Variables · Addition of Means · Addition of Variances	
Chapter 2	Statistical Methods	43
2.1	Introduction	43
2.2	Point Estimation of Parameters	47
	Maximum Likelihood Method	

2.3	Confidence Intervals	51
2.4	Testing of Hypotheses and Making Decisions	58
	Errors in Tests	
2.5	Quality Control	65
	Control Chart for the Mean · Control Chart for the Variance · Control Chart for the Standard Deviation · Control Chart for the Range	
2.6	Acceptance Sampling	70
	Errors in Acceptance Sampling · Rectification	
2.7	Goodness of Fit	74

Part II Introduction to Machine Learning

Acronyms	83
-----------------	-----------

Supplemental List of Items

List of Figures

1.1	The histogram of the data given in Exercise 1	5
1.2	Examples of Venn diagrams.	9
1.3	A visual comparison of the Stirling formula and the actual values of the factorial function.	19
1.4	A visual representation of the Eq. <i>extendedeqref : neq : dist</i> – 9.	23
1.5	Binomial distribution with different values of probability with a sample size of 50. .	29
1.6	The Poisson distribution with different mean (μ) values.	30
1.7	The probability density distribution of hyper-geometric distribution with different parameters.	31
1.8	The poster child of probability and statistics, the normal distribution.	32
1.9	A visual representation between the relationship of PDF and CDF.	33
1.10	Many samples from a bivariate normal distribution. The marginal distributions are shown on the z-axis. The marginal distribution of X is also approximated by creating a histogram of the X coordinates without consideration of the Y coordinates. . .	37
2.1	Original map by John Snow showing the clusters of cholera cases in the London epidemic of 1854, drawn and lithographed by Charles Cheffins.	46
2.2	The student-t distribution with different degrees of freedom m	55
2.3	Chi-square distribution with different degrees of freedom.	56
2.4	The t -distribution used in example. As can be seen, anything left of the critical line would tell us to reject the hypothesis, whereas if the t value lies on the Right Hand Side (RHS), then the test would tell us the null hypothesis is true.	59
2.5	Illustration of Type I and II errors in testing a hypothesis $\theta = \theta_0$ against an alternative $\theta = \theta_0$	63
2.6	Control charts for the mean (upper part of figure) and the standard deviation in the case of the samples on p. 1089	65
2.7	OC curve of the sampling plan $n = 2$ and $c = 0$ for lots of size $N = 20$	71
2.8	The OC curve with Poisson process applied to it.	71
2.9	OC curve and AOQ curve for the sampling plan.	73

List of Tables

1.1	Stem and Leaf plot of the data given in Example 1.	4
-----	--	---

2.1	Useful c values based on a given confidence (γ) value.	52
2.2	Twelve samples of five (5) values each (diameter of small cylinders, measured in mm)	66
2.3	Acceptance Sampling and Hypothesis Testing	72

List of Examples

1.1	Recording Data	4
1.2	Leaf Plots	4
1.3	Histogram	5
1.4	Empirical Rule Outliers and z-Score	7
1.5	Sample Spaces of Random Experiments & Events	7
1.6	Fair Die	10
1.7	Coin Tossing	12
1.8	Mutually Exclusive Events	12
1.9	Union of Arbitrary Events	13
1.10	Multiplication Rule	13
1.11	Sampling w/o Replacement	15
1.12	An Encrypted Message	17
1.13	Sampling Light-bulbs	18
1.14	Waiting Time Problem	22
1.15	Continuous Distribution	23
1.16	Mean and Variance	25
1.17	Poisson Distribution	30
1.18	Marginal Distributions of a Discrete Two-Dimensional Random Variable	38
1.19	Independence and Dependence	39
2.1	Maximum Likelihood of Gaussian Distribution	49
2.2	Maximum Likelihood of Poisson Distribution	49
2.3	For Science	50
2.4	Sampling the Population	50
2.5	Confidence Interval for Mean with known Variance in Normal Distribution	52
2.6	Sample Size Needed for a Confidence Interval of Prescribed Length	54
2.7	Confidence Interval for Mean of Normal Distribution with Unknown Variance	55
2.8	Test of a Hypothesis	58
2.9	Test for the Mean of the Normal Distribution with Known Variance	63
2.10	Printed Circuit Boards	75

List of Theorems

1.1	First Definition of Probability	10
1.2	General Definition of Probability	11
1.3	Complementation Rule	12

1.4	Addition Rule for Mutually Exclusive Events	12
1.5	Addition Rule for Arbitrary Events	12
1.6	Multiplication Rule	13
1.7	Permutations	16
1.8	Permutations	17
1.9	Combinations	18
1.10	Random Variable	20
1.11	Mean of a Symmetric Distribution	26
1.12	Transformation of Mean and Variance	26
1.13	Relationship between PDF and CDF	33
1.14	Normal Probabilities for Intervals	33
1.15	Limit Theorem of De Moivre and Laplace	34
1.16	Addition of Means	41
1.17	Multiplication of Means	41
1.18	Addition of Variances	42
2.1	Sum of Independent Normal Random Variables	52
2.2	Student's t-Distribution	55
2.3	Chi-Square Distribution	56
2.4	Chi-square Test for $F(x)$ being the Distribution Function of a Population . .	75



Part I

Probability & Statistics

Part Contents

Chapter 1	Theory of Probability	3
Chapter 2	Statistical Methods	43

If you torture the data enough, nature will always confess.

(Ronald Coase (1982). How should economists choose?. American Enterprise Institute.)

Chapter 1

Theory of Probability

Table of Contents

1.1	Introduction	3
1.2	Probability	10
1.3	Permutations & Combinations	16
1.4	Random Variables and Probability Distributions	20
1.5	Mean and Variance of a Distribution	25
1.6	Binomial, Poisson, and Hyper-geometric Distributions	28
1.7	Normal Distribution	32
1.8	Distribution of Several Random Variables	36

1.1 Introduction

When the data we are working are influenced by “chance”, by factors whose effect we cannot predict exactly¹, we have to rely on **probability theory**. The application of this theory nowadays appears in numerous fields such as from studying a game of cards to the global financial market and allow us to model processes of chance called **random experiments**.

¹This could be weather data, stock prices, life spans or ties, etc.

In such an experiment we observe a **random variable** X , that is, a function whose values in a trial² occur “by chance” according to a **probability distribution** which gives the individual probabilities, which possible values of X may occur in the long run.

²a performance of an experiment.

i.e., each of the six (6) faces of a die should occur with the same probability, $1/6$.

Or we may simultaneously observe more than one random variable, for instance, height and weight of persons or hardness and tensile strength of steel.

But enough about spoiling all the fun and let's begin with looking at data.

1.1.1 Representing Data

Data can be represented numerically or graphically in different ways

i.e., a news website may contain tables of stock prices and currency exchange rates, curves or bar charts illustrating economical or political developments, or pie charts showing how inflation is calculated.

And there are numerous other representations of data for special purposes. In this section, we will discuss the use of standard representations of data in statistics³.

³There are various software dedicated to analyse and visualise statistical data. Some of these include: R, a statistical programming language, Python, MATLAB, ...

Example 1.1: Recording Data

Sample values, such as observations and measurements, should be recorded in the order in which they occur. Sorting, that is, ordering the sample values by size, is done as a first step of investigating properties of the sample and graphing it. As an example let's look at super alloys.

Super alloys is a collective name for alloys used in jet engines and rocket motors, requiring high temperature,⁴ high strength, and excellent resistance to oxidation. Thirty (30) specimens of Hastelloy C (nickel-based steel, investment cast) had the tensile strength (in 1000 lb>sq in.), recorded in the order obtained and rounded to integer values.

89	77	88	91	88	93	99	79	87	84	86	82	88	89	78	(1.1)
90	91	81	90	83	83	92	87	89	86	89	81	87	84	89	

Of course depending on the need the data needs to be sorted which is shown below:

77	78	79	81	81	82	83	83	84	84	86	86	87	87	87
88	88	88	89	89	89	89	89	90	90	91	91	92	93	99

⁴typically 1000 °C.

1.1.2 Graphic Representation of Data

Let's now use the data we have seen in Example 1 and see the methods we can use for graphic representations.

Example 1.2: Leaf Plots

One of the simplest yet most useful representations of data [1]. For Eq. (1.1) it is shown in **Table 1.1**.

LO	12 12
7	789
8	1123344
8	6677788899999
9	001123
9	9
HI	172

Table 1.1: Stem and Leaf plot of the data given in Example 1.



The numbers in Eq. (1.1) range from 78 to 99; which you can also see this in the sorted list. To visualise this data feature, we divide these numbers into five (5) groups:

75-79, 80-84, 85-89, 90-94, 95-99.

The integers in the tens position of the groups are 7, 8, 8, 9, 9. These form the stem which can be seen in **Table**

The first leaf is 789, representing 77, 78, 79. The second leaf is 1123344, representing 81, 81, 82, 83, 83, 84, 84. And so on. The number of times a value occurs is called its **absolute frequency**.

Therefore in this example, 78 has absolute frequency 1, the value 89 has absolute frequency 5, etc.



Example 1.3: Histogram

For large sets of data, histograms are better in displaying the distribution of data than stem-and-leaf plots. The principle is explained in **Fig. 1.1**.

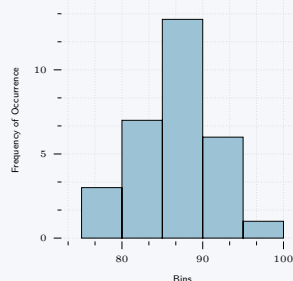


Figure 1.1: The histogram of the data given in Exercise 1.

The bases of the rectangles in seen in **Fig. 1.1** are the x -intervals⁵ where there range is:

74.5 – 79.5, 79.5 – 84.5, 84.5 – 89.5,
89.5 – 94.5, 94.5 – 99.5,

whose midpoints, known as **class marks**, are

$x = 77, 82, 87, 92, 97,$

respectively. The height of a rectangle with class mark x is the relative class frequency $f_{\text{rel}}(x)$, defined as the number of data values in that class interval, divided by n ($= 30$ in our case). Hence the areas of the rectangles are proportional to these relative frequencies,

0.10, 0.23, 0.43, 0.17, 0.07,

so that histograms give a good impression of the distribution of data.

⁵known as class intervals.

Mean, Standard Deviation, and Variance

Medians and quartiles are easily obtained by ordering and counting.⁶

⁶This can be done without the need of calculators.

However this method does **NOT** give full information on data as you can change data values to some extent without changing the median.

The average size of the data values can be measured in a more refined way by the **mean**:

Definition of Arithmetic Mean

$$\bar{x} \equiv \frac{1}{n} \sum_{j=1}^n x_j = \frac{1}{n} (x_1 + x_2 + \cdots + x_n).$$

This is the **arithmetic mean** of the data values, obtained by taking their sum and dividing by the data size (n). Therefore the arithmetic mean for values given in Eq. (1.1) is:

$$\bar{x} = \frac{1}{30} (89 + 77 + \cdots + 89) = \frac{260}{3} \approx 86.7 \quad \blacksquare$$

As we can see every data value contributes, and changing one of them will change the mean. Similarly, the spread⁷ of the data values can be measured in a more refined way by the **standard deviation** (s) or by its square, the variance⁸

⁷also known as variability.

⁸The symbol for variance is interesting as each domain have their own definition, as s^2 , σ^2 and $\text{Var}()$ are all acceptable symbols.

Definition of Variance

$$s^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2 = \frac{1}{n-1} [(x_1 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2]$$

Therefore, to obtain the variance of the data, take the difference⁹ of each data value from the mean, square it, take the sum of these n squares, and divide it by $n - 1$.

⁹i.e., $x_j - \bar{x}$.

To get the standard deviation s , take the square root of s^2 .

Returning back to our super alloy example, using $\bar{x} = 260/3$ ¹⁰, we get for the data given in Eq. (1.1) the variance:

¹⁰which we calculated previously

$$s^2 = \frac{1}{29} \left[\left(89 - \frac{260}{3}\right)^2 + \left(77 - \frac{260}{3}\right)^2 + \cdots + \left(89 - \frac{260}{3}\right)^2 \right] = \frac{2006}{87} \approx 23.06 \quad \blacksquare$$

Therefore, the standard deviation is calculated to be:

$$s = \sqrt{2006/87} \approx 4.802$$

The standard deviation has the same dimension as the data values,¹¹ which is an advantage, whereas, the variance is preferable to the standard deviation in developing statistical methods.

¹¹i.e., if your data set contains measurements in cm its standard deviation would be cm whereas its variance is cm^2 .

Information: Empirical Rule

For any round-shaped symmetric distribution of data the intervals:

$$\bar{x} \pm s, \quad \bar{x} \pm 2s, \quad \bar{x} \pm 3s, \quad \text{contain about} \quad 68\%, \quad 95\%, \quad 99.7\%.$$

respectively, of the data points. This information is quite useful in doing quick calculation of statistical properties such as the quality of production which will be the focus in Chapter ??.

**Example 1.4:** Empirical Rule Outliers and z-Score

For the data set given in Example 1.1, with $\bar{x} = 86.7$ and $s = 4.8$, the three (3) intervals in the Rule are:

$$81.9 \leq x \leq 91.5, \quad 77.1 \leq x \leq 96.3, \quad 72.3 \leq x \leq 101.1$$

and contain 73% (22 values remain, 5 are too small, and 5 too large), 93% (28 values, 1 too small, and 1 too large), and 100%, respectively.

If we reduce the sample by omitting the outlier value of 99, mean and standard deviation reduce to $\bar{x}_{\text{red}} = 86.2$, and $s_{\text{red}} = 4.3$, approximately, and the percentage values become 67% (5 and 5 values outside), 93% (1 and 1 outside), and 100%.

Finally, the relative position of a value x in a set of mean $\bar{x}$ and standard deviation s can be measured by the **z-score**:

$$z(s) = \frac{x - \bar{x}}{s}$$

This is the distance of x from the mean $\bar{x}$ measured in multiples of s . For instance:

$$z(s) = \frac{(83 - 86.7)}{4.8} = -0.77$$

This is negative because 83 lies below the mean. By the empirical rule, the extreme z-values are about -3 and 3. ■

1.1.3 Experiments & Outcomes

Now we have the basis covered, it is time to look at probability theory.¹² This theory has the purpose of providing mathematical models of situations affected or even governed by **change effects**, for instance, in weather forecasting, life insurance, quality of technical products,¹³ traffic problems, and, of course, games of chance with cards or dice, where the accuracy of these models can be tested by suitable observations or experiments. Let's start by defining some standard terms:

¹²Sometimes known as probability calculus.

¹³computers, batteries, steel sheets, etc.

experiment A process of measurement or observation, in a laboratory, in a factory, . . .

randomness Situation where absolute prediction is **NOT** possible.

trial A single performance of an experiment.

outcome The result of a trial.¹⁴

¹⁴also known as sample point.

sample space Defined as S , the set of all possible outcomes of an experiment.

Example 1.5: Sample Spaces of Random Experiments & Events

■ Inspecting a lightbulb | $S = \{\text{Defective, Non-defective}\}$.

■ Rolling a die | $S = \{1, 2, 3, 4, 5, 6\}$ with it's events

$$A = \{1, 3, 5\} \text{ ("Odd number")} \quad \text{and} \quad B = \{2, 4, 6\} \text{ ("Even number")}, \text{ etc.}$$



■ Counting daily traffic accidents in Vienna | $S = \{\text{the integers in some interval}\}$.

1.1.4 Unions, Intersections, and Complements of Events

In connection with basic probability laws we also need the following concepts and facts about events¹⁵ $A, B, C, \dots$ of a given sample space S .

¹⁵ called subsets of the probability event S .

- The **union** $A \cup B$ of A and B consists of all points in A or B or both.
- The **intersection** $A \cap B$ of A and B consists of all points that are in both A and B .

If A and B have **no** points in common, we write

$$A \cap B = \emptyset$$

where $\emptyset$ is the empty set¹⁶ and we call A and B **mutually exclusive** (or **disjoint**) as, in a trial, the occurrence of A *excludes* that of B (and conversely)—if your die turns up an odd number, it cannot turn up an even number in the same trial, or a coin cannot turn up Head (H) and Tail (T) at the same time.

¹⁶ This means it is a set which contains nothing.

- The **Complement** of A is A^c ¹⁷. This is the set of all the points of S *not* in A . Therefore,

$$A \cap A^c = \emptyset, \quad A \cup A^c = S.$$

¹⁷ Another notation for the complement of A is $\bar{A}$ (instead of A^c), but we shall not use this because in set theory $\bar{A}$ is used to denote the *closure* of A .

Unions and intersections of more events are defined similarly. The **union**:

$$\bigcup_{j=1}^m A_j = A_1 \cup A_2 \cup \dots \cup A_m.$$

of events $A_1, \dots, A_m$ consists of all points that are in at least one A_j . Similarly for the union $A_1 \cup A_2 \cup \dots$ of infinitely many subsets $A_1, A_2, \dots$ of an *infinite* sample space S (that is, S consists of infinitely many points). The **intersection**:

$$\bigcap_{j=1}^m A_j = A_1 \cap A_2 \cap \dots \cap A_m$$

of $A_1, \dots, A_m$ consists of the points of S that are in each of these events. Similarly for the intersection $A_1 \cap A_2 \cap \dots$ of infinitely many subsets of S .

Working with events can be illustrated and facilitated by **Venn diagrams** for showing unions, intersections, and complements, as in **Fig. 1.2**, which are typical examples expressing the concept covered previously.

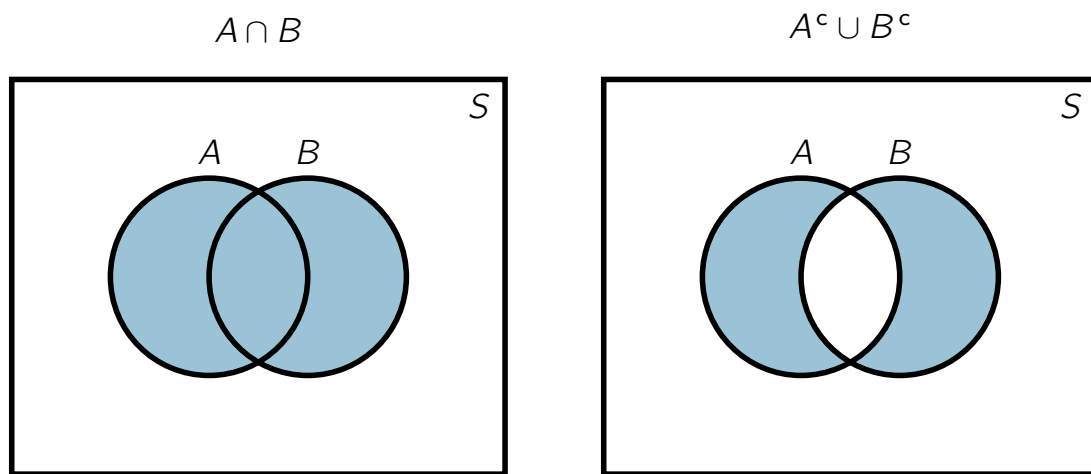


Figure 1.2
Examples of Venn
diagrams.

1.2 Probability

The **probability** of an event A in an experiment is to measure **how frequently** A is roughly to occur if we make many trials. If we flip a coin, then heads H and tails T will appear **about** equally¹⁸ often.

¹⁸on the condition, the measurements are done for a long time.

we say that H and T are **"equally likely."**

Similarly, for a regularly shaped die of homogeneous material¹⁹ each of the six (6) outcomes $1, \dots, 6$ will be equally likely. These are examples of experiments in which the sample space S consists of finitely many outcomes (points) that for reasons of some symmetry can be regarded as equally likely.

¹⁹called a fair dice

Let's formulate this in a theory.

Theory 1.1: First Definition of Probability

If the sample space S of an experiment consists of **finitely** many outcomes (points) being equally likely, the probability $P(A)$ of an event A is defined to be:

$$P(A) = \frac{\text{Number of points in } A}{\text{Number of points in } S}.$$

From this definition it follows immediately, in particular, the probability of all events occurring in the sample space S is:

$$P(S) = 1.$$

Example 1.6: Fair Die

In rolling a fair die once:

1. What is the probability $P(A)$ of A of obtaining a 5 or a 6?
2. The probability of B : "Even number"?

Solution

The six outcomes are equally likely, so that each has probability $1/6$. Therefore:

$$P(A) = \frac{2}{6} = \frac{1}{3} \quad \text{and} \quad P(B) = \frac{3}{6} = \frac{1}{2} \quad \blacksquare$$

The above theory takes care of many games as well as some practical applications, but not of all experiments, as in many problems we do not have finitely many equally likely outcomes. To arrive at a more general definition of probability, we regard probability as the counterpart of **relative frequency**:

$$f_{\text{rel}}(A) = \frac{f(A)}{n} = \frac{\text{Number of times } A \text{ occurs}}{\text{Number of trials}} \quad (1.2)$$

Now if A did not occur, then $f(A) = 0$. If A always occurred, then $f(A) = n$. These are of course extreme cases. Division by n gives:

$$0 \leq f_{\text{rel}}(A) \leq 1$$

In particular, for $A = S$ we have $f(S) = n$ as S always occurs²⁰. Division by n gives:

$$f_{\text{rel}}(S) = 1$$

Finally, if A and B are **mutually exclusive**, they cannot occur together. Therefore the absolute frequency of their union $A \cup B$ must equal the sum of the absolute frequencies of A and B . Division by n gives the same relation for the relative frequencies:

$$f_{\text{rel}}(A \cup B) = f_{\text{rel}}(A) + f_{\text{rel}}(B)$$

We can now extend the definition of probability to experiments in which equally likely outcomes are not available.

Theory 1.2: General Definition of Probability

Given a sample space S , with each event A of S (A being a subset of S) there is associated a number $P(A)$, called the **probability** of A , such the following **axioms of probability** are satisfied.

- For every A in S ,

$$0 \leq P(A) \leq 1.$$

- The entire sample space S has the probability

$$P(S) = 1.$$

- For **mutually exclusive** events A and B :

$$P(A \cup B) = P(A) + P(B) \quad (A \cap B = \emptyset).$$

- If S is infinite²¹, the previous statement has to be replaced by Eq. (1.2), where for mutually exclusive events $A_1, A_2, \dots$,

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots.$$

²⁰ meaning that some event always occurs

²¹i.e., has infinitely many points.

In the infinite case the subsets of S on which $P(A)$ is defined are restricted to form a so-called σ -algebra.

Basic Theorems of Probability

We will see that the axioms of probability will enable us to build up probability theory and its application to statistics. We begin with three (3) basic theorems. The first one is useful if we can get the probability of the complement A^c more easily than $P(A)$ itself.

Theory 1.3: Complementation Rule

For an event A and its complement A^c in a sample space S ,

$$P(A^c) = 1 - P(A)$$

Example 1.7: Coin Tossing

Five (5) coins are tossed simultaneously.

Find the probability of the event A :

At least one head turns up. Assume that the coins are fair.

As each coin can turn up either heads or tails, the sample space consists of $2^5 = 32$ outcomes. Given the coins are fair, we may assign the same probability ($1/32$) to each outcome. Then the event A^c (No heads turn up) consists of only 1 outcome. Hence $P(A^c) = 1/32$, and the answer is:

$$P(A) = 1 - P(A^c) = \frac{31}{32} \quad \blacksquare$$

Theory 1.4: Addition Rule for Mutually Exclusive Events

For **mutually exclusive events** $A_1, \dots, A_m$ in a sample space S ,

$$P(A_1 \cup A_2 \cup \dots \cup A_m) = P(A_1) + P(A_2) + \dots + P(A_m).$$

Example 1.8: Mutually Exclusive Events

If the probability that on any workday a garage will get 10-20, 21-30, 31-40, over 40 cars to service is 0.20, 0.35, 0.25, 0.12, respectively, what is the probability that on a given workday the garage gets at least 21 cars to service?

Solution

As these are mutually exclusive events, the answer is:

$$0.35 + 0.25 + 0.12 = 0.72 \quad \blacksquare$$

However, most situations, events will **NOT** be mutually exclusive. Then we have the following theorem to formalise the previous statement.

Theory 1.5: Addition Rule for Arbitrary Events

For events A and B in a sample space, their union is defined as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

For **mutually exclusive** events A and B we have $A \cap B = \emptyset$ by definition:

$$P(\emptyset) = 0$$

Example 1.9: Union of Arbitrary Events

In tossing a fair die, what is the probability of getting an odd number or a number less than 4?

Solution

Let A be the event "Odd number" and B the event "Number less than 4." As these event are linked we can write:

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{2}{3}$$

as $A \cup B = \text{Odd number less than 4} = \{1, 3\}$ ■

Conditional Probability and Independent Events

It is often required to find the probability of an event B given the condition of an event A occurs. This probability is called the **conditional probability** of B given A and is denoted by $P(B|A)$.

In this case A serves as a new, reduced, sample space, and that probability is the fraction of $P(A)$ which corresponds to $A \cap B$. Therefore,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where} \quad P(B) \neq 0 \quad (1.3)$$

Similarly, the conditional probability of A given B is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{where} \quad P(A) \neq 0$$

Theory 1.6: Multiplication Rule

Given A and B are events defined in a sample space S and $P(A) \neq 0, P(B) \neq 0$, then

$$P(A \cap B) = P(A) P(B|A) = P(B) P(A|B). \quad (1.4)$$

Example 1.10: Multiplication Rule

In producing screws, let:

- A mean "screw too slim",
- B mean "screw too short."



Let $P(A) = 0.1$ and let the conditional probability that a slim screw is also too short be $P(B|A) = 0.2$. What is the probability that a screw that we pick randomly from the lot produced will be both too slim and too short?

Solution

$$P(A \cap B) = P(A) P(B|A) = 0.1 \times 0.2 = 0.02 = 2\% \quad \blacksquare$$

Independent Events

If events A and B are such that

$$P(A \cap B) = P(A) P(B),$$

they are called **independent events**. Assuming $P(A) \neq 0, P(B) \neq 0$, we see from Eq. (1.3) - Eq. (1.4):

$$P(A|B) = P(A), \quad P(B|A) = P(B).$$

This means that the probability of A does not depend on the occurrence or nonoccurrence of B , and conversely. This justifies the term **independent**.

Independence of m Events

Similarly, m events $A_1, \dots, A_m$ are called **independent** if:

$$P(A_1 \cap \dots \cap A_m) = P(A_1) \dots P(A_m)$$

as well as for every k different events $A_{j_1}, A_{j_2}, \dots, A_{j_k}$.

$$P(A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_k}) = P(A_{j_1}) P(A_{j_2}) \dots P(A_{j_k})$$

where $k = 2, 3, \dots, m-1$. Accordingly, three events A, B, C are independent if and only if

$$\begin{aligned} P(A \cap B) &= P(A) P(B), \\ P(B \cap C) &= P(B) P(C), \\ P(C \cap A) &= P(C) P(A), \\ P(A \cap B \cap C) &= P(A) P(B) P(C). \end{aligned}$$

Sampling

Our next example has to do with randomly drawing objects, *one at a time*, from a given set of objects. This is called **sampling from a population**, and there are two ways of sampling, as follows.

- **In sampling with replacement**, the object that was drawn at random is placed back to the given set and the set is mixed thoroughly. Then we draw the next object at random.
- **In sampling without replacement** the object that was drawn is put aside.

Example 1.11: Sampling w/o Replacement

A box contains 10 screws, three (3) of which are defective. Two screws are drawn at random. Find the probability that neither of the two screws is defective.

Solution

We consider the events

- A** First drawn screw non-defective,
- B** Second drawn screw non-defective.

We can see:

$$P(A) = \frac{1}{10}$$

as 7 of the 10 screws are non-defective and we sample at random, so that each screw has the same probability ($\frac{1}{10}$) of being picked.

If we sample with replacement, the situation before the second drawing is the same as at the beginning, and $P(B) = \frac{7}{10}$. The events are independent, and the answer is

$$P(A \cap B) = P(A) P(B) = 0.7 \cdot 0.7 = 0.49\%.$$

If we sample without replacement, then $P(A) = \frac{7}{10}$, as before. If **A** has occurred, then there are 9 screws left in the box, 3 of which are defective.

Thus $P(B|A) = \frac{6}{9} = \frac{2}{3}$, therefore:

$$P(A \cap B) = \frac{7}{10} \cdot \frac{2}{3} = 47\% \quad \blacksquare$$

1.3 Permutations & Combinations

Permutations and combinations help in finding probabilities $P(A) = a/k$ by systematically counting the number a of points of which an event A consists.

where, k is the number of points of the sample space S .

The practical difficulty is that a may often be surprisingly large, so that actual counting becomes hopeless. For example, if in assembling some instrument you need 10 different screws in a certain order and you want to draw them randomly from a box²² the probability of obtaining them in the required order is only $1/3,628,800$ because there are exactly:

$$10! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 = 3,628,800$$

orders in which they can be drawn. Similarly, in many other situations the numbers of orders, arrangements, etc. are often incredibly large.

²²Of course, this goes without saying, there is nothing but screws in this imaginary box.

1.3.1 Permutations

A **permutation** of given things²³ is an arrangement of these things in a row in some order.

²³such as elements or objects.

i.e., for three (3) letters a, b, c there are $3! = 1 \cdot 2 \cdot 3 = 6$ permutations: abc, acb, bca, cab, cba

Let's write this behaviour down as a theory:

Theory 1.7: Permutations

Different things

The number of permutations of n different things taken all at a time is

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n. \quad (1.5)$$

Classes of Equal Things

If n given things can be divided into c classes of alike things differing from class to class, then the number of permutations of these things taken all at a time is

$$\frac{n!}{n_1! n_2! \dots n_c!} \quad \text{where} \quad n_1 + n_2 + \dots + n_c = n,$$

where n_j is the number of things in the j^{th} class.

Permutation of n things taken k at a time

A permutation containing only k of the n given things. Two such permutations consisting of the same k elements, in a different order, are different, by definition.

i.e., there are 6 different permutations of the three letters a, b, c , taken two letters at a time, ab, ac, bc, ba, ca, cb .

Permutation of n things taken k at a time with repetitions

An arrangement obtained by putting any given thing in the first position, any given thing, including a repetition of the one just used, in the second, and continuing until k positions are filled.

i.e., there are $3^2 = 9$ different such permutations of a, b, c taken 2 letters at a time, namely, the preceding 6 permutations and aa, bb, cc .

Theory 1.8: Permutations

The number of different permutations of n different things taken k at a time **without repetitions** is

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!},$$

and **with repetitions** is,

$$n^k.$$

Example 1.12: An Encrypted Message

In an encrypted message the letters are arranged in groups of five (5) letters, called words. Knowing the letter can be repeated, we see that the number of different such words is

$$26^5 = 11,881,376 \quad \blacksquare$$

For the case of different such words containing each letter no more than once is

$$\frac{26!}{(26-5)!} = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 = 7,893,600 \quad \blacksquare$$

1.3.2 Combinations

In a permutation, the **order of the selected things is essential**. In contrast, a **combination** of a given things means any selection of one or more things **without regard to order**. There are two (2) kinds of combinations, as follows:

1. The number of **combinations of n different things, taken k at a time, without repetitions** is the number of sets that can be made up from the n given things, each set containing k different things and no two (2) sets containing exactly the same k things.
2. The number of **combinations of n different things, taken k at a time, with repetitions** is the number of sets that can be made up of k things chosen from the given n things, each

being used as often as desired.

i.e, there are three (3) combinations of the three (3) letters a, b, c , taken two (2) letters at a time, without repetitions, namely, ab, ac, bc , and six such combinations with repetitions, namely, ab, ac, bc, ca, bb, cc .

Theory 1.9: Combinations

The number of different combinations of n different things taken, k at a time, **without repetitions**, is:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1) \cdots (n-k+1)}{1 \cdot 2 \cdots k},$$

and the number of those combinations **with repetitions** is:

$$\binom{n+k-1}{k}. \quad (1.6)$$

Example 1.13: Sampling Light-bulbs

The number of samples of five (5) light-bulbs that can be selected from a lot of 500 bulbs is

$$\binom{500}{5} = \frac{500!}{5!495!} = \frac{500 \cdot 499 \cdot 498 \cdot 497 \cdot 476}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = 255,244,687,600 \quad \blacksquare$$

1.3.3 Factorial Function

In Eq. (1.5)-Eq. (1.6) the **factorial function** is relatively straightforward. By definition²⁴,

$$0! = 1.$$

Values may be computed recursively from given values by

$$(n+1)! = (n+1)n!.$$

For large n the function is very large and hard to keep track of. A convenient approximation for large n is the **Stirling formula**, defined as:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{where} \quad e = 2.718 \cdots \quad (1.7)$$

where $\sim$ is read asymptotically equal²⁵ and means that the ratio of the two sides of Eq. (1.7) approaches 1 as n approaches infinity.

²⁴This is done by convention. An intuitive way to look at it is $n!$ counts the number of ways to arrange n distinct objects in a line, and there is only one way to arrange nothing.

²⁵it means the percentage difference between the vertical distances between points on the two graphs approaches 0.

1.3.4 Binomial Coefficients

The **binomial coefficients** are defined by the following formula:

$$\binom{a}{k} = \frac{(a)(a-1)(a-2) \cdots (a-k+1)}{k!} \quad \text{where} \quad (k \geq 0, \text{integer}) \quad (1.8)$$

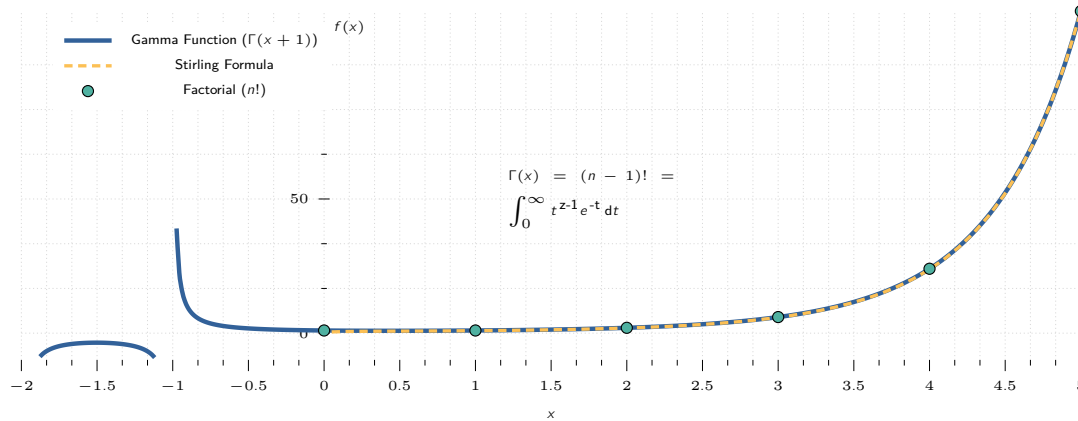


Figure 1.3: A visual comparison of the Stirling formula and the actual values of the factorial function.

The numerator has k factors. Furthermore, we define

$$\binom{a}{0} = 1, \quad \text{in particular,} \quad \binom{0}{0} = 1.$$

For integer $a = n$ we obtain from Eq. (1.8):

$$\binom{n}{k} = \binom{n}{n-k} \quad (n \geq 0 \text{ and } 0 \leq k \leq n).$$

Binomial coefficients may be computed recursively, because

$$\binom{a}{k} + \binom{a}{k+1} = \binom{a+1}{k+1} \quad (k \geq 0, \text{ integer}).$$

Formula Eq. (1.8) also gives:

$$\binom{-m}{k} = (-1)^k \binom{m+k-1}{k} \quad \text{where } k \geq 0, \text{ integer and } m > 0.$$

There are two (2)important relations worth mentioning:

$$\sum_{s=0}^{n-1} \binom{k+s}{k} = \binom{n+k}{k+1} \quad (k \geq 0 \text{ and } n \geq 1)$$

and

$$\sum_{k=0}^r \binom{p}{k} \binom{q}{r-k} = \binom{p+q}{r} \quad (r \geq 0, \text{ integer}).$$

1.4 Random Variables and Probability Distributions

In the beginning of this chapter we considered frequency distributions of data²⁶. These distributions show the **absolute** or **relative** frequency of the data values.

²⁶Remember we did a histogram and a stem-and-leaf plot.

Similarly, a **probability distribution** or, a **distribution**, shows the probabilities of events in an experiment. The quantity we observe in an experiment will be denoted by X and called a random variable²⁷ as the value it will assume in the next trial depends on the **stochastic process**

²⁷or **stochastic variable** if you want to be pedantic.

i.e., if you roll a die, you get one of the numbers from 1 to 6, but you don't know which one will show up next. An example would be, $X = \text{Number a die turns up}$, which is a random variable.

If we count²⁸, we have a **discrete random variable and distribution**. If we **measure** (electric voltage, rainfall, hardness of steel), we have a **continuous random variable and distribution**. For both cases (discrete, discontinuous), the distribution of X is determined by the **distribution function**:

²⁸cars on a road, defective parts in a production, tosses until a die shows the first six (6).

$$F(x) = P(X \leq x) \quad (1.9)$$

This is the probability that in a trial, X will assume any value not exceeding x .

The terminology is unfortunately **NOT** uniform across the field as $F(x)$ is sometimes also called the **cumulative distribution function**.

For Eq. (1.9) to make sense in both the discrete and the continuous case we formulate conditions as follows.

Theory 1.10: Random Variable

A **random variable** X is a function defined on the sample space S of an experiment. Its values are real numbers. For every number a the probability:

$$P(X = a),$$

with which X assumes a is defined. Similarly, for any interval I , the probability

$$P(X \in I),$$

with which X assumes any value in I is defined²⁹.

²⁹Although this definition is very general, in practice only a very small number of distributions will occur over and over again in applications.

From Eq. (1.9) we can define the fundamental formula for the probability corresponding to an interval $a < x \leq b$:

$$P(a < X \leq b) = F(b) - F(a). \quad (1.10)$$

This follows because $X \leq a$ (X assumes any value **NOT** exceeding a) and $a < X \leq b$ (X assumes



any value in the interval $a < x \leq b$ are **mutually exclusive** events, so based on Eq. (1.9):

$$\begin{aligned} F(b) &= P(X \leq b) = P(X \leq a) + P(a < X \leq b) \\ &= F(a) + P(a < X \leq b) \end{aligned}$$

and subtraction of $F(a)$ on both sides gives Eq. (1.10).

1.4.1 Discrete Random Variables and Distributions

By definition, a random variable X and its distribution are **discrete** if X assumes only **finitely** many or at most countably many values $x_1, x_2, x_3, \dots$, called the **possible values** of X , with positive probabilities,

$$p_1 = P(X = x_1), p_2 = P(X = x_2), p_3 = P(X = x_3), \dots$$

whereas the probability $P(X \in I)$ is zero for any interval I containing no possible value. Clearly, the discrete distribution of X is also determined by the **probability function** $f(x)$ of X , defined by

$$f(x) = \begin{cases} p_j & \text{if } x = x_j \\ 0 & \text{otherwise} \end{cases} \quad \text{where } j = 1, 2, \dots,$$

From this we get the values of the **distribution function** $F(x)$ by taking sums,

$$F(x) = \sum_{x_j \leq x} f(x_j) = \sum_{x_j \leq x} p_j \quad (1.11)$$

where for any given x we sum all the probabilities p_j for which x_j is smaller than or equal to that of x . This is a **step function** with upward jumps of size p_j at the possible values x_j of X and constant in between. The two (2) useful formulas for discrete distributions are readily obtained as follows. For the probability corresponding to intervals we have from Eq. (1.10) and Eq. (1.11):

$$P(a < X \leq b) = F(b) - F(a) = \sum_{a < x_j \leq b} p_j \quad (1.12)$$

This is the sum of all probabilities p_j for which x_j satisfies $a < x_j \leq b$ ³⁰. From this and $P(S) = 1$ we obtain the following formula.

³⁰Be careful about $<$ and $\leq$ as the former means it is **NOT** included and the latter means it is.

$$\sum_j p_j = 1 \quad (\text{sum of all probabilities}). \quad (1.13)$$

Example 1.14: Waiting Time Problem

In tossing a fair coin, let X be the Number of trials until the first head appears. Then, by independence of events we get (where H is heads, and T is tails):

$$\begin{aligned}P(X = 1) &= P(H) = \frac{1}{2} \\P(X = 2) &= P(TH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \\P(X = 3) &= P(TTH) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}\end{aligned}$$

and in general, $P(X = n) = \left(\frac{1}{2}\right)^n$, $n = 1, 2, 3, \dots$ which when all possible event are summed up will always give 1.

1.4.2 Continuous Random Variables and Distributions

Discrete random variables appear in experiments in which we count³¹. Continuous random variables appear in experiments in which we **measure** (lengths of screws, voltage in a power line, etc.). By definition, a random variable X and its distribution are of *continuous type* or, briefly, **continuous**, if its distribution function $F(x)$, defined in Eq. (1.9), can be given by an integral³²:

³¹defectives in a production, days of sunshine in Kufstein, customers in a line, etc.

³²we write v as a toss-away variable because x is needed as the upper limit of the integral.

$$F(x) = \int_{-\infty}^x f(v) \, dv \quad (1.14)$$

whose integrand $f(x)$, called the **density** of the distribution, is **non-negative**, and is continuous, perhaps except for finitely many x -values. Differentiation gives the relation of f to F as

$$f(x) = F'(x)$$

for every x at which $f(x)$ is continuous.

From Eq. (1.10) and Eq. (1.14) we obtain the very important formula for the probability corresponding to an interval³³:

³³This is an analog of Eq. (1.12)

$$P(a < X \leq b) = F(b) - F(a) = \int_a^b f(v) \, dv \quad (1.15)$$

Which can be seen visually in **Fig. 1.4**. From Eq. (1.14) and $P(S) = 1$ we also have the analogue of Eq. (1.13):

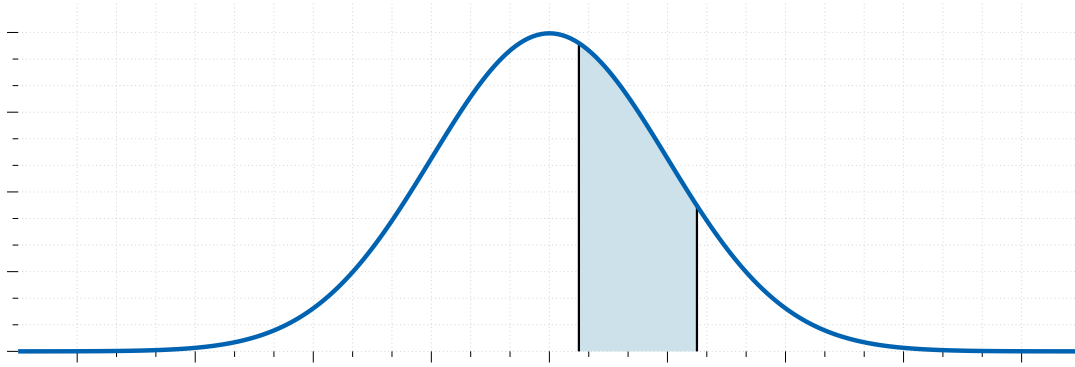


Figure 1.4: A visual representation of the Eq. (1.15).

$$\int_{-\infty}^{\infty} f(v) dv = 1.$$

Continuous random variables are **simpler than discrete ones** with respect to intervals as, in the continuous case the four probabilities corresponding to $a < X \leq b$, $a < X < b$, $a \leq X \leq b$, and $a \leq X < b$ with any fixed a and b ($> a$) are all the same.

The next example illustrates notations and typical applications of our present formulas.

Example 1.15: Continuous Distribution

Let X have the density function:

$$f(x) = 0.75(1 - x^2) \quad \text{if} \quad -1 \leq x \leq 1,$$

and zero otherwise. Find:

1. The distribution function.
2. Find the probabilities $P(-\frac{1}{2} \leq X \leq \frac{1}{2})$ and $P(\frac{1}{2} \leq X \leq 2)$
3. Find x such that $P(X \leq x) = 0.95$.

Solution

From Eq. (1.14), we obtain $F(x) = 0$ if $x \leq -1$,

$$F(x) = 0.75 \int_{-1}^x (1 - v^2) dv = 0.5 + 0.75x - 0.25x^3 \quad \text{if} \quad -1 < x \leq 1,$$

and $F(x) = 1$ if $x > 1$. From this and Eq. (1.15) we get:

$$P(-\frac{1}{2} \leq X \leq \frac{1}{2}) = F(\frac{1}{2}) - F(-\frac{1}{2}) = 0.75 \int_{-1/2}^{1/2} (1 - v^2) dv = 68.75\%$$

because $P(-\frac{1}{2} \leq X \leq \frac{1}{2}) = P(-\frac{1}{2} < X \leq \frac{1}{2})$ for a continuous distribution we can write:

$$P(\frac{1}{4} \leq X \leq 2) = F(2) - F(\frac{1}{4}) = 0.75 \int_{1/4}^1 (1 - v^2) dv = 31.64\%.$$

Note that the upper limit of integration is 1, not 2. Finally,

$$P(X \leq x) = F(x) = 0.5 + 0.75x - 0.25x^2 = 0.95.$$

Algebraic simplification gives $3x - x^3 = 1.8$. A solution is $x = 0.73$, approximately ■



1.5 Mean and Variance of a Distribution

The mean μ and variance σ^2 of a random variable X and of its distribution are the theoretical counterparts of the mean $\bar{x}$ and variance s^2 of a frequency distribution and serve a similar purpose.

The mean characterises the central location and the variance the spread (the variability) of the distribution. The **mean** μ is defined by:

$$(a) \quad \mu = \sum_j x_j f(x_j) \quad (\text{Discrete distribution}) \quad (1.16a)$$

$$(b) \quad \mu = \int_{-\infty}^{\infty} x f(x) dx \quad (\text{Continuous distribution}) \quad (1.16b)$$

and the **variance** σ^2 by:

$$(a) \quad \sigma^2 = \sum_j (x_j - \mu)^2 f(x_j) \quad (\text{Discrete distribution}) \quad (1.17a)$$

$$(b) \quad \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad (\text{Continuous distribution}) \quad (1.17b)$$

σ (the positive square root of σ^2) is called the standard deviation³⁴ of X and its distribution. f is the probability function or the density, respectively, in (a) and (b).

³⁴Sometimes it is known as $\text{Var}(x)$

The mean μ is also denoted by $E(X)$ and is called the **expectation of** X because it gives the average value of X to be expected in many trials.

Quantities such as μ and σ^2 that measure certain properties of a distribution are called **parameters**. μ and σ^2 are the two (2) most important ones.

From Eq. (1.17a) and Eq. (1.17b), we see that³⁵:

$$\sigma^2 > 0$$

³⁵except for a discrete distribution with only one possible value.

We assume that μ and σ^2 exist³⁶, as is the case for practically all distributions that are useful in applications.

³⁶and finite.

Example 1.16: Mean and Variance

The random variable X , *Number of heads in a single toss of a fair coin*, has the possible values $X = 0$ and $X = 1$ with probabilities $P(X = 0) = \frac{1}{2}$ and $P(X = 1) = \frac{1}{2}$. From Eq. (1.16a) we thus obtain the mean:

$$\mu = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2},$$

and Eq. (1.17a) gives the variance:

$$\sigma^2 = (0 - \frac{1}{2})^2 \cdot \frac{1}{2} + (1 - \frac{1}{2})^2 \cdot \frac{1}{2} = \frac{1}{4} \quad \blacksquare$$



Symmetry

We can obtain the mean μ without calculation if a distribution is symmetric. Indeed, we can write:

Theory 1.11: Mean of a Symmetric Distribution

If a distribution is **symmetric** with respect to $x = c$, that is,

$$f(c - x) = f(c + x)$$

then $\mu = c$.

Transformation of Mean and Variance

Given a random variable X with mean μ and variance σ^2 , we want to calculate the mean and variance of $X^* = a_1 + a_2X$, where a_1 and a_2 are given constants.

This problem is important in statistics, where it often appears.

Theory 1.12: Transformation of Mean and Variance

If a random variable X has mean μ and variance σ^2 , then the random variable:

$$X^* = a_1 + a_2X \quad \text{where} \quad a_2 > 0$$

has the mean μ^* and variance σ^{*2} , where

$$\mu^* = a_1 + a_2\mu \quad \text{and} \quad \sigma^{*2} = a_2^2\sigma^2.$$

In particular, the **standardised random variable** Z corresponding to X , given by:

$$Z = \frac{X - \mu}{\sigma}$$

has the mean 0 and the variance 1.

Expectation & Moments

If we recall, Eq. (1.16a) and Eq. (1.16b) define the mean of X^{37} , written $\mu = E(X)$. More generally, if $g(x)$ is **non-constant** and continuous for all x , then $g(X)$ is a random variable. Therefore its **mathematical expectation** or, briefly, its expectation $E(g(X))$ is the value of $g(X)$ to be expected on the average, defined by:

³⁷the value of X to be expected on the average

$$E(g(X)) = \sum_j g(x_j) f(x_j) \quad \text{or} \quad E(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

In the formula on the Left Hand Side (LHS), f is the probability function of the discrete random variable X . In the formula on the RHS, f is the density of the continuous random variable X .

Important special cases are the k^{th} of X (where $k = 1, 2, \dots$)

$$E(X^k) = \sum_j x_j^k f(x_j) \quad \text{or} \quad \int_{-\infty}^{\infty} x^k f(x) dx$$

and the k^{th} of X ($k = 1, 2, \dots$)

$$E([X - \mu]^k) = \sum_j (x_j - \mu)^k f(x_j) \quad \text{or} \quad \int_{-\infty}^{\infty} (x - \mu)^k f(x) dx.$$

This includes the first moment, the **mean** of X

$$\mu = E(X) \quad \text{where} \quad k = 1$$

It also includes the second central moment, the **variance** of X

$$\sigma^2 = E([X - \mu]^2) \quad \text{where} \quad k = 2.$$



1.6 Binomial, Poisson, and Hyper-geometric Distributions

These are the three (3) most important **discrete** distributions, with numerous applications therefore are worth of a bit of a detailed look.

Of course these are not the only distributions present. There are as many distributions as there are problems with some distributions used in wide variety of fields (Gaussian) whereas some are used only in a very narrow field (Nakagami).

Binomial Distribution

The **binomial distribution** occurs in problems involving of chance³⁸.

What we are interested is in the number of times an event A occurs in n **independent** trials. In each trial, the event A has the same probability $P(A) = p$. Then in a trial, A will **NOT** occur with probability $q = 1 - p$. In n trials the random variable that interests us is:

$X = \text{Number of times the event } A \text{ occurs in } n \text{ trials.}$

X can assume the values $0, 1, \dots, n$, and we want to determine the corresponding probabilities. Now $X = x$ means that A occurs in x trials and in $n - x$ trials it does not occur. We can write this down as follows:

$$\underbrace{A \ A \ \dots \ A}_{x \text{ times}} \quad \text{and} \quad \underbrace{B \ B \ \dots \ B}_{n - x \text{ times}} \quad (1.18)$$

Here $B = A^c$ is the complement of A , meaning that A does not occur. We now use the assumption that the trials are independent³⁹. Hence Eq. (1.18) has the probability:

$$\underbrace{p \ p \ \dots \ p}_{x \text{ times}} \cdot \underbrace{q \ q \ \dots \ q}_{n - x \text{ times}} = p^x q^{n-x} \quad (1.19)$$

Now Eq. (1.18) is just one order of arranging x A 's and $n - x$ B 's. We will now calculate the number of permutations of n things⁴⁰ consisting of two (2) classes;

1. class 1 containing the $n_1 = x$ A 's
2. class 2 containing the $n - n_1 = n - x$ B 's

This number is:

$$\frac{n!}{x!(n - x)!} = \binom{n}{x}.$$

³⁸rolling a dice, quality inspection (e.g., counting of the number of defectives), opinion plots (counting number of employees favouring certain schedule changes, etc.), medicine (e.g., recording the number of patterns who covered on a new medication)

³⁹e.g., they do **NOT** influence each other

⁴⁰the n outcomes of the n trials

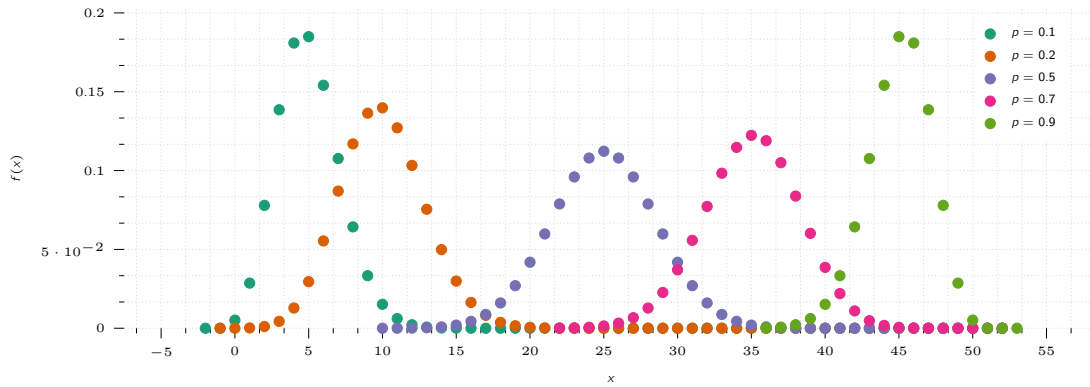


Figure 1.5: Binomial distribution with different values of probability with a sample size of 50.

Accordingly, Eq. (1.19), multiplied by this binomial coefficient, gives the probability $P(X = x)$ of $X = x$, that is, of obtaining A precisely x times in n trials. Hence X has the probability function:

$$f(x) = \binom{n}{x} p^x q^{n-x} \quad (x = 0, 1, \dots, n) \quad (1.20)$$

and $f(x) = 0$ otherwise. The distribution of X with probability function (2) is called the **binomial distribution** or *Bernoulli distribution*. The occurrence of A is called success⁴¹ and the non-occurrence of A is called *failure*.

⁴¹regardless of what it actually is; it may mean that you miss your plane or lose your watch

The mean and variance of the binomial distribution is:

$$\mu = np \quad \text{and} \quad \sigma^2 = npq$$

For the *symmetric* case of equal chance of success and failure ($p = q = \frac{1}{2}$) this gives the mean $n/2$, the variance $n/4$, and the probability function

$$f(x) = \binom{n}{x} \left(\frac{1}{2}\right)^n \quad (x = 0, 1, \dots, n).$$

Poisson Distribution

The discrete distribution with infinitely many possible values and probability function:

$$f(x) = \frac{\mu^x}{x!} e^{-\mu} \quad \text{where} \quad x = 0, 1, \dots \quad (1.21)$$

is called the **Poisson distribution**, named after *S. D. Poisson*. Fig. 1.6 shows Eq. (1.21) for some values of μ ⁴².

⁴²While μ is used here, some textbook use λ

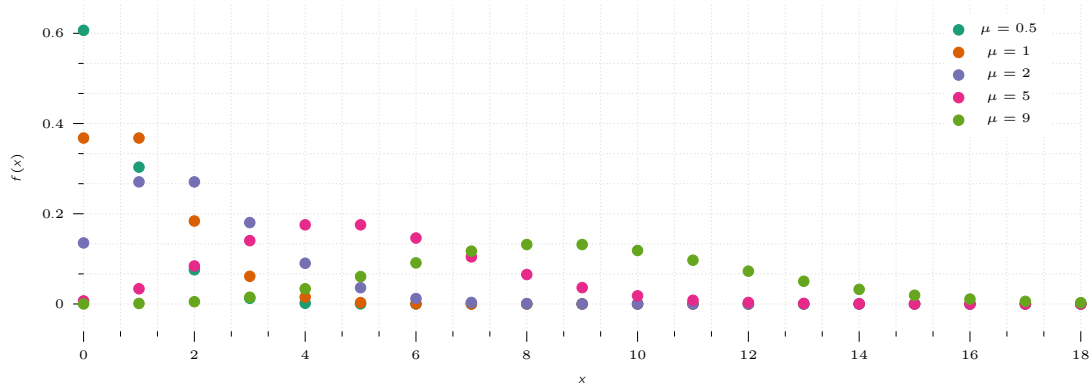


Figure 1.6: The Poisson distribution with different mean (μ) values.

It can be proved that this distribution is obtained as a limiting case of the binomial distribution, if we let $p \rightarrow 0$ and $n \rightarrow \infty$ so that the mean $\mu = np$ approaches a finite value. The Poisson distribution has the mean μ and the variance:

$$\sigma^2 = \mu. \quad (1.22)$$

Fig. 1.6 gives the impression that, with increasing mean, the spread of the distribution increases, thereby illustrating formula Eq. (1.22), and that the distribution becomes more and more symmetric.⁴³

⁴³approximately

Example 1.17: Poisson Distribution

If the probability of producing a defective screw is $p = 0.01$, what is the probability that a lot of 100 screws will contain more than 2 defectives?

Solution

The complementary event is A^c . No more than 2 defectives. For its probability we get, from the binomial distribution with mean $\mu = np = 1$, the value.

$$P(A^c) = \binom{100}{0} 0.99^{100} + \binom{100}{1} 0.01 \cdot 0.99^{100} + \binom{100}{2} 0.01^2 \cdot 0.99^{100}.$$

Since p is very small, we can approximate this by the much more convenient Poisson distribution with mean $\mu = np = 100 \cdot 0.01 = 1$, obtaining.

$$P(A^c) = e^{-1} \left(1 + 1 + \frac{1}{2} \right) = 91.97\%.$$

Thus $P(A) = 8.03\%$. Show that the binomial distribution gives $P(A) = 7.94\%$, so that the Poisson approximation is quite good ■

1.6.1 Sampling with Replacement

This means that we draw things from a given set one by one, and after each trial we replace the thing drawn⁴⁴ before we draw the next thing. This guarantees **independence of trials** and leads to the **binomial distribution**. Indeed, if a box contains N things, for example, screws, M of which are defective, the probability of drawing a defective screw in a trial is $p = M/N$. Hence the probability of drawing a nondefective screw is $q = 1 - p = 1 - M/N$, and Eq. (1.20) gives the probability of drawing x defectives in n trials in the form:

$$f(x) = \binom{n}{x} \left(\frac{M}{N}\right)^x \left(1 - \frac{M}{N}\right)^{n-x} \quad (x = 0, 1, \dots, n). \quad (1.23)$$

⁴⁴put it back to the given set and mix.

1.6.2 Sampling without Replacement: Hyper-geometric Distribution

Sampling without replacement means that we return no screw to the box. Then we no longer have independence of trials, and instead of Eq. (1.23) the probability of drawing x defectives in n trials is:

$$f(x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad \text{where} \quad x = 1, 2, \dots, n.$$

The distribution with this probability function is called the hyper-geometric distribution⁴⁵.

The hypergeometric distribution has the mean and the variance:

$$\mu = n \frac{M}{N} \quad \text{and} \quad \sigma^2 = \frac{nM(N-M)(N-n)}{N^2(N-1)}.$$

⁴⁵because its moment generating function can be expressed by the hypergeometric function, which is a fact only useful to write it in a margin.

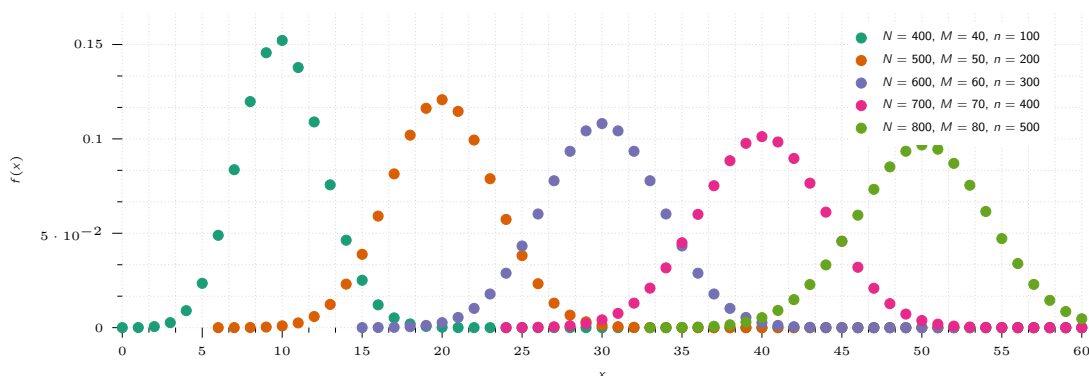


Figure 1.7: The probability density distribution of hyper-geometric distribution with different parameters.

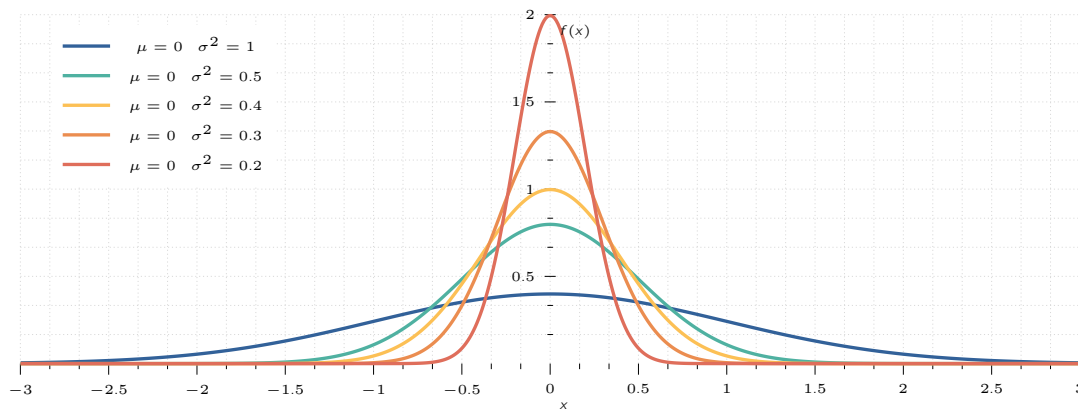


Figure 1.8: The poster child of probability and statistics, the normal distribution.

1.7 Normal Distribution

Turning from discrete to continuous distributions, in this section we discuss the normal distribution. This is the most important continuous distribution because in applications many random variables are normal random variables⁴⁶ or they are approximately normal or can be transformed into normal random variables in a relatively simple fashion. Furthermore, the normal distribution is a useful approximation of more complicated distributions, and it also occurs in the proofs of various statistical tests.

⁴⁶that is, they have a normal distribution.

The **normal distribution** or *Gauss distribution* is defined as the distribution with the density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \quad (1.24)$$

where $\exp$ is the exponential function with base $e = 2.718\ldots$. This is simpler than it may at first look. $f(x)$ has these features (see also **Fig. 1.8**).

1. μ is the mean, and σ the standard deviation.
2. $1/(\sigma\sqrt{2\pi})$ is a constant factor that makes the area under the curve of $f(x)$ from $-\infty$ to ∞ equal to 1, as it must be⁴⁷.
3. The curve of $f(x)$ is symmetric with respect to $x = \mu$ because the exponent is **quadratic**. Hence for $\mu = 0$ it is symmetric with respect to the y -axis $x = 0$ ⁴⁸.
4. The exponential function in Eq. (1.24) goes to zero very fast—the faster the smaller the standard deviation σ is, as it should be, as seen in **Fig. 1.8**.

⁴⁷Having a probability higher than 1 does NOT make sense

⁴⁸This distribution is also known as bell-shaped curves.

1.7.1 Distribution Function

From Eq. (1.23) and Eq. (1.24) we see that the normal distribution has the **distribution function** of the following form:

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x \exp\left[-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2\right] dv. \quad (1.25)$$

Here we needed x as the upper limit of integration and wrote v (instead of x) in the integrand.

For the corresponding **standardised normal distribution** with mean 0 and standard deviation 1 we denote $F(x)$ by $\Phi(z)$. Then we simply have from Eq. (1.25).

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du. \quad (1.26)$$

This integral cannot be integrated by one of the methods of calculus.

But this is no serious handicap because its values can be obtained from standardised tables. These values are needed in working with the normal distribution. The curve of $\Phi(z)$ is S-shaped. It increases monotone from 0 to 1 and intersects the vertical axis at $\frac{1}{2}$, as shown in Fig. 1.9.

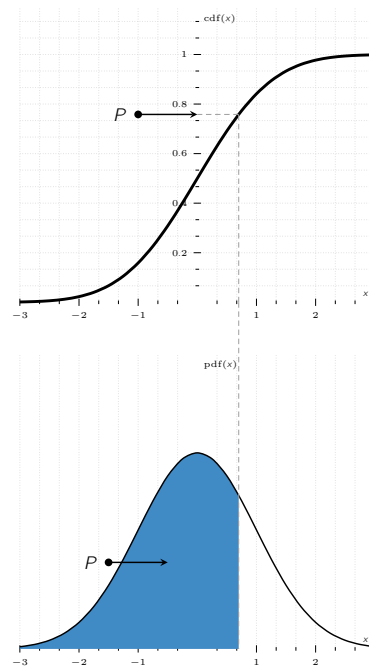


Figure 1.9: A visual representation between the relationship of PDF and CDF.

Theory 1.13: Relationship between PDF and CDF

The distribution function $F(x)$ of the normal distribution with any μ and σ is related to the standardised distribution function $\Phi(z)$ in Eq. (1.26) by the formula

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

Theory 1.14: Normal Probabilities for Intervals

The probability a normal random variable X with mean μ and standard deviation σ assume any value in an interval $a < x \leq b$ is:

$$P(a < X \leq b) = F(b) - F(a) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$



1.7.2 Numeric Values

In practical work with the normal distribution it is good to remember that about 67% of all values of X to be observed will be between $\mu \pm \sigma$, about 95% between $\mu \pm 2\sigma$, and practically all between the **three-sigma limits** $\mu \pm 3\sigma$:

$$\begin{aligned}P(\mu - \sigma < X \leq \mu + \sigma) &\approx 68\% \\P(\mu - 2\sigma < X \leq \mu + 2\sigma) &\approx 95.5\% \\P(\mu - 3\sigma < X \leq \mu + 3\sigma) &\approx 99.7\%.\end{aligned}$$

The aforementioned formulas show that a value deviating from μ by more than σ , 2σ , or 3σ will occur in one of about 3, 20, and 300 trials, respectively.

In tests⁴⁹, we shall ask, conversely, for the intervals that correspond to certain given probabilities; practically most important use the probabilities of 95%, 99%, and 99.9%. For these, the answers are $\mu \pm 2\sigma$, $\mu \pm 2.6\sigma$, and $\mu \pm 3.3\sigma$, respectively.

⁴⁹Which we shall cover in Chapter ??.

More precisely,

$$\begin{aligned}P(\mu - 1.96\sigma < X \leq \mu + 1.96\sigma) &\approx 95\% \\P(\mu - 2.58\sigma < X \leq \mu + 2.58\sigma) &\approx 99\% \\P(\mu - 3.29\sigma < X \leq \mu + 3.29\sigma) &\approx 99.9\%.\end{aligned}$$

1.7.3 Normal Approximation of the Binomial Distribution

The probability function of the binomial distribution, as a reminder, is:

$$f(x) = \binom{n}{x} p^x q^{n-x} \quad (x = 0, 1, \dots, n). \quad (1.29)$$

If n is large, the binomial coefficients and powers become very inconvenient. It is of great practical⁵⁰ importance that, in this case, the normal distribution provides a good approximation of the binomial distribution, according to the following theorem, one of the most important theorems in all probability theory.

⁵⁰and theoretical

Theory 1.15: Limit Theorem of De Moivre and Laplace

For large n ,

$$f(x) \sim f^*(x) \quad \text{where} \quad x = 0, 1, \dots, n$$

Here f is given by Eq. (1.29). The function

$$f^*(\cdot) = \frac{1}{\sqrt{2\pi npq}} \exp\left(-\frac{z^2}{2}\right), \quad \text{and} \quad z = \frac{x - np}{\sqrt{npq}}$$

is the density of the normal distribution with mean $\mu = np$ and variance $\sigma^2 = npq$ (the mean and variance of the binomial distribution). Furthermore, for any nonnegative integers a and b ($b > a$):

$$P(a \leq X \leq b) = \sum_{x=a}^b \binom{n}{x} p^x q^{n-x} \sim \Phi(\beta) - \Phi(\alpha)$$

where,

$$\alpha = \frac{a - np - 0.5}{\sqrt{npq}} \quad \text{and} \quad \beta = \frac{b - np + 0.5}{\sqrt{npq}}$$



1.8 Distribution of Several Random Variables

Distributions of two (2) or more random variables are of interest for two (2) reasons:

1. They occur in experiments in which we observe several random variables, for example, carbon content X and hardness Y of steel, amount of fertiliser X and yield of corn Y , height X_1 , weight X_2 , and blood pressure X_3 of persons, and so on.
2. They will be needed in the mathematical justification of the methods of statistics in Chapter ??.

In this section we consider two (2) random variables X and Y or, as we also say, a **two-dimensional random variable** (X, Y) . For (X, Y) the outcome of a trial is a pair of numbers $X = x$, $Y = y$, briefly $(X, Y) = (x, y)$, which we can plot as a point in the XY -plane.

The **two-dimensional probability distribution** of the random variable (X, Y) is given by the **distribution function**

$$F(x, y) = P(X \leq x, Y \leq y).$$

This is the probability that in a trial, X will assume any value not greater than x and in the same trial, Y will assume any value not greater than y . $F(x, y)$ determines the probability distribution **uniquely**, because extending the analogy we developed previously, $P(a < X \leq b) = F(b) - F(a)$, we now have for a rectangle defined using the following equation:

$$P(a_1 < X \leq b_1, a_2 < Y \leq b_2) = F(b_1, b_2) - F(a_1, b_2) - F(b_1, a_2) + F(a_1, a_2).$$

As before, in the two-dimensional case we shall also have **discrete** and **continuous** random variables and distributions.

1.8.1 Discrete Two-Dimensional Distribution

In analogy to the case of a single random variable, we call (X, Y) and its distribution **discrete** if (X, Y) can assume only finitely many or at most countably infinitely many pairs of values (x_1, y_1) , $(x_2, y_2), \dots$ with positive probabilities, whereas the probability for any domain containing none of those values of (X, Y) is zero.

Let (x_i, y_i) be any of those values and let $P(X = x_i, Y = y_j) = p_{ij}$ (where we admit that p_{ij} may be 0 for certain pairs of subscripts i). Then we define the **probability function** $f(x, y)$ of (X, Y) by:

$$f(x, y) = p_{ij} \quad \text{if} \quad x = x_i, y = y_j \quad \text{and} \quad f(x, y) = 0 \quad \text{otherwise;}$$

where, $i = 1, 2, \dots$ and $j = 1, 2, \dots$ independently. In analogy to Eq. (1.11), we now have for the

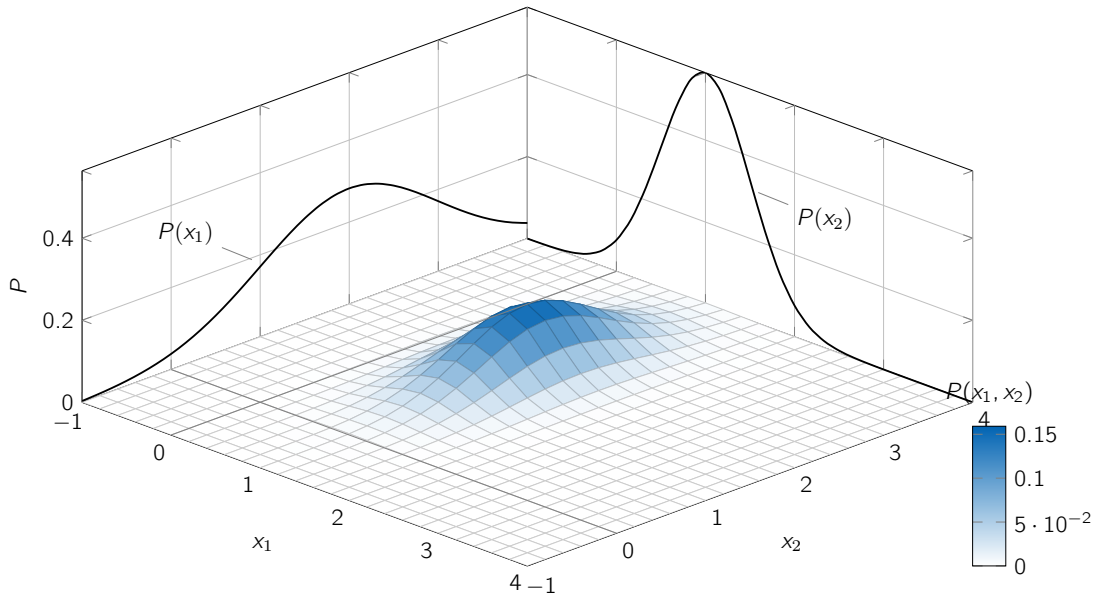


Figure 1.10: Many samples from a bivariate normal distribution. The marginal distributions are shown on the z-axis. The marginal distribution of X is also approximated by creating a histogram of the X coordinates without consideration of the Y coordinates.

distribution function the formula:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} f(x_i, y_j).$$

Instead of Eq. (1.13), we now have the condition:

$$\sum_i \sum_j f(x_i, y_j) = 1.$$

1.8.2 Continuous Two-Dimensional Distribution

In analogy to the case of a single random variable, we call (X, Y) and its distribution **continuous** if the corresponding distribution function $F(x, y)$ can be given by a double integral:

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(x^*, y^*) dx^* dy^* \quad (1.30)$$

whose integrand f , called the **density** of (X, Y) , is non-negative everywhere, and is continuous, possibly except on finitely many curves.

From Eq. (1.30) we obtain the probability that (X, Y) assume any value in a rectangle (Fig. 523) given by the formula:

$$P(a_1 < X \leq b_1, a_2 < Y \leq b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x, y) dx dy$$



1.8.3 Marginal Distributions of a Discrete Distribution

This is a rather natural idea, without counterpart for a single random variable.

It amounts to being interested only in one of the two variables in (X, Y) , say, X , and asking for its distribution, called the **marginal distribution** of X in (X, Y) . So we ask for the probability $P(X = x, Y \text{ arbitrary})$.

Since (X, Y) is discrete, so is X . We get its probability function, call it $f_1(x)$, from the probability function $f(x, y)$ of (X, Y) by summing over y :

$$f_1(x) = P(X = x, Y, \text{arbitrary}) = \sum_y f(x, y) \quad (1.31)$$

where we sum all the values of $f(x, y)$ that are not 0 for that x .

From Eq. (1.31) we see that the distribution function of the marginal distribution of X is

$$F_1(x) = P(X \leq x, Y, \text{arbitrary}) = \sum_{x^* \leq x} f_1(x^*).$$

Similarly, the probability function

$$f_2(y) = P(X \text{ arbitrary}, Y \equiv y) = \sum_x f(x, y)$$

determines the **marginal distribution** of Y in (X, Y) . Here we sum all the values of $f(x, y)$ that are not zero for the corresponding y . The distribution function of this marginal distribution is

$$F_2(y) = P(X \text{ arbitrary}, Y \equiv y) = \sum_{y^* \equiv y} f_2(y^*).$$

Example 1.18: Marginal Distributions of a Discrete Two-Dimensional Random Variable

In drawing 3 cards with replacement from a bridge deck let us consider

(X, Y) where X = Number of queens and Y = Number of kings or aces.

The deck has 52 cards. These include 4 queens, 4 kings, and 4 aces. Therefore, in a single trial a queen has probability:

$$\frac{4}{52} = \frac{1}{13}$$

and a king or ace:

$$\frac{8}{52} = \frac{2}{13}$$

This gives the probability function of (X, Y) as:

$$f(x, y) = \frac{3!}{x!y!(3-x-y)} \left(\frac{1}{13}\right)^x \left(\frac{2}{13}\right)^y \left(\frac{10}{13}\right)^{3-x-y} \quad \text{where } (x+y \leq 3)$$

and $f(x, y) = 0$ otherwise.



1.8.4 Marginal Distributions of a Continuous Distribution

This is conceptually the same as for discrete distributions, with probability functions and sums replaced by densities and integrals. For a continuous random variable (X, Y) with density $f(x, y)$ we now have the **marginal distribution** of X in (X, Y) , defined by the distribution function

$$F_1(x) = P(X \leq x, -\infty < Y < \infty) = \int_{-\infty}^x f_1(x^*) dx^*$$

with the density f_1 of X obtained from $f(x, y)$ by integration over y ,

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

Interchanging the roles of X and Y , we obtain the **marginal distribution** of Y in (X, Y) with the distribution function

$$F_2(y) = P(-\infty < X < \infty, Y \leq y) = \int_{-\infty}^y f_2(y^*) dy^*$$

and density

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

1.8.5 Independence of Random Variables

X and Y in a, discrete or continuous, random variable (X, Y) are said to be **independent** if

$$F(x, y) = F_1(x)F_2(y)$$

holds for all (x, y) . Otherwise these random variables are said to be **dependent**. Necessary and sufficient for independence is

$$f(x, y) = f_1(x)f_2(y)$$

for all x and y . Here the f 's are the above probability functions if (X, Y) is discrete or those densities if (X, Y) is continuous.

Example 1.19: Independence and Dependence

In tossing a 50 cent and a 20 cent coin, with X being the number of heads on the 50 cent, and Y number of heads on the 20 cent, we may assume the values 0 or 1 and are independent.

Extension of Independence to n -Dimensional Random Variables. This will be needed throughout Chapter ???. The distribution of such a random variable $\mathbf{X} = (X_1, \dots, X_n)$ is determined by a **distribution function** of the form

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n).$$



The random variables $X_1, \dots, X_n$ are said to be **independent** if

$$F(x_1, \dots, x_n) = F_1(x_1)F_2(x_2) \cdots F_n(x_n)$$

for all $(x_1, \dots, x_n)$. Here $F_j(x_j)$ is the distribution function of the marginal distribution of X_j in $\mathbf{X}$, that is,

$$F_j(x_j) = P(X_j \leq x_j, X_k \text{ arbitrary}, k = 1, \dots, n, k \neq j).$$

Otherwise these random variables are said to be **dependent**.

1.8.6 Functions of Random Variables

When $n = 2$, we write $X_1 = X$, $X_2 = Y$, $x_1 = x$, $x_2 = y$. Taking a non-constant continuous function $g(x, y)$ defined for all x, y , we obtain a random variable $Z = g(X, Y)$.

For example, if we roll two (2) dice and X and Y are the numbers the dice turn up in a trial, then $Z = X + Y$ is the sum of those two (2) numbers.

In the case of a **discrete** random variable (X, Y) we may obtain the probability function $f(z)$ of $Z = g(X, Y)$ by summing all $f(x, y)$ for which $g(x, y)$ equals the value of z considered; thus

$$f(z) = P(Z = z) = \sum_{g(x,y)=z} \sum f(x, y).$$

Hence the distribution function of Z is

$$F(z) = P(Z \leq z) = \sum_{g(x,y) \leq z} \sum f(x, y),$$

where we sum all values of $f(x, y)$ for which $g(x, y) \leq z$.

In the case of a **continuous** random variable (X, Y) we similarly have

$$F(z) = P(Z \leq z) = \iint_{g(x,y) \leq z} f(x, y) \, dx \, dy$$

where for each z we integrate the density $f(x, y)$ of (X, Y) over the region $g(x, y) \leq z$ in the xy -plane, the boundary curve of this region being $g(x, y) = z$.

1.8.7 Addition of Means

The number

$$E(g(X, Y)) = \begin{cases} \sum_x \sum_y g(x, y) f(x, y) & \text{where } X, Y \text{ are discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx \, dy & \text{where } X, Y \text{ are continuous} \end{cases} \quad (1.32)$$



is called the **mathematical expectation** or, briefly, the **expectation of** $g(X, Y)$. Here it is assumed that the double series converges absolutely and the integral of $|g(x, y)|/(x, y)$ over the y -plane exists.⁵¹ Since summation and integration are linear processes, we have from Eq. (1.32):

⁵¹meaning it is finite.

$$E(ag(X, Y) + bh(X, Y)) = aE(g(X, Y)) + bE(h(X, Y))$$

An important special case is

$$E(X + Y) = E(X) + E(Y),$$

and by induction we have the following result.

Theory 1.16: Addition of Means

The mean (expectation) of a sum of random variables equals the sum of the means (expectations), that is,

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n).$$

We can also deduce the following statement:

Theory 1.17: Multiplication of Means

The mean (expectation) of the product of independent random variables equals the product of the means (expectations), that is,

$$E(X_1 X_2 \cdots X_n) = E(X_1) E(X_2) \cdots E(X_n).$$

and in the continuous case the proof of the relation is similar⁵².

⁵²This is left as an exercise to the reader.

1.8.8 Addition of Variances

A final matter to cover is how we can sum up variances. Similar to before, let $Z = X + Y$ and denote the mean and variance of Z by μ and σ^2 .

Then we first have:

$$\sigma^2 = E([Z - \mu]^2) = E(Z^2) - [E(Z)]^2$$

From (24) we see that the first term on the right equals

$$E(Z^2) = E(X^2 + 2XY + Y^2) = E(X^2) + 2E(XY) + E(Y^2).$$

For the second term on the right we obtain from Theorem 1

$$[E(Z)]^2 = [E(X) + E(Y)]^2 = [E(X)]^2 + 2E(X)E(Y) + [E(Y)]^2$$

By substituting these expressions into the formula for σ^2 we have

$$\begin{aligned} \sigma^2 &= E(X^2) - [E(X)]^2 + E(Y^2) - [E(Y)]^2 \\ &\quad + 2[E(XY) - E(X)E(Y)]. \end{aligned}$$

the expression in the first line on the right is the sum of the variances of X and Y , which we denote by σ_1^2 and σ_2^2 , respectively.

The quantity in the second line (except for the factor 2) is:

$$\sigma_{XY} = E(XY) - E(X)E(Y),$$

and is called the **covariance** of X and Y . Consequently, our result is

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 + 2\sigma_{XY}.$$

If X and Y are **independent**, then

$$E(XY) = E(X)E(Y);$$

hence $\sigma_{XY} = 0$, and

$$\sigma^2 = \sigma_1^2 + \sigma_2^2$$

Extension to more than two variables gives the basic

Theory 1.18: Addition of Variances

The variance of the sum of independent random variables equals the sum of the variances of these variables.

Chapter 2

Statistical Methods

Table of Contents

2.1	Introduction	43
2.2	Point Estimation of Parameters	47
2.3	Confidence Intervals	51
2.4	Testing of Hypotheses and Making Decisions	58
2.5	Quality Control	65
2.6	Acceptance Sampling	70
2.7	Goodness of Fit	74

2.1 Introduction

Statistical¹ methods consists of a wide range of tools for designing and evaluating random experiments to **obtain information** about practical problems:

¹The word is derived from New Latin *statistica* or *statisticus* ("of the state")

such as exploring the relation between iron content and density of iron ore, the quality of raw material or manufactured products, the efficiency of air-conditioning systems, the performance of certain cars, the effect of advertising, the reactions of consumers to a new product, etc.

Therefore, it is an important topic for any engineer as **Random variables** occur more frequently in engineering² than one would think. For example, properties of mass-produced articles³ always exhibit **random variation**, due to small⁴ differences in raw material or manufacturing processes.

²and of course elsewhere.

³such as screws, light bulbs, electric machines, etc.

⁴often uncontrollable

Therefore, the diameter of screws is a random variable X and we have **non-defective screws**, with diameter between given tolerance limits, and **defective screws**, with diameter outside those limits.

We can ask for the distribution of X , for the percentage of defective screws to be expected, and for necessary improvements of the production process.

Samples are selected from populations:

20 screws from 1000 screws, 100 of 5000 voters, 8 behaviours in a wildlife observation.

as inspecting the entire sample, would be expensive, time-consuming, impossible or even senseless.⁵

To obtain a meaningful sense of information, samples must be **random selections**. Each of the 1000 screws must have the same chance of being sampled,⁶ at least approximately. Only then will the sample mean:

$$\bar{x} = \frac{1}{20} (x_1 + \cdots x_{20}) \quad \text{where} \quad n = 20,$$

will be a **good approximation** of the population mean μ , and the accuracy of the approximation will generally improve with increasing n , as we shall see.

This is also applicable to other statistical quantities such as standard deviation, variance, etc.

Independent sample values will be obtained in experiments with an infinite sample space S certainly for the **normal distribution**. This is also true in sampling with replacement. It is approximately true in drawing **small samples** from a large finite population.⁷ However, if we sample without replacement from a small population, the effect of dependence of sample values may be considerable.

Random numbers help in obtaining samples that are in fact random selections. This is sometimes not easy to accomplish as there are numerous subtle factors which can bias sampling.⁸ Random numbers can be obtained from a **random number generator**

It is important to state that the numbers generated by a computer are **NOT** truly random, as are calculated by a tricky formula that produces numbers that do have practically all the essential features of true randomness. Because these numbers eventually repeat, they must not be used in cryptography, for example, where true randomness is required.

Information: Generating Random Numbers

To select a sample of size $n = 10$ from 80 given ball bearings, we number the bearings from 1 to 80. We then let the generator randomly produce 10 of the integers from 1 to 80 and include the bearings with the numbers obtained in our sample, for example,

44 55 57 03 61 51 68 22 34 77

or whichever number pops up in your head.⁹

Representing and processing data were considered in the previous chapter in connection with **frequency distributions**. These are the **empirical counterparts** of probability distributions and helped motivating axioms and properties in probability theory. The new aspect in this chapter is **randomness**:

⁵It would be inconceivable for a company who produces over a billion light bulbs to test all their products. That is why we have return policies.

⁶of being drawn when we sample.

⁷for instance, 5 or 10 of 1000 items.

⁸Such as by personal interviews, by poorly working machines, by the choice of non-typical observation conditions, etc.

⁹Of course in a professional setting you can't just write numbers like that as there is also a pattern when we make successive random number. Before the prevalence of computers there used to be books containing random numbers which people consulted.

i.e., the data are samples selected **randomly** from a population.

Accordingly, we can already use the plots we have used in probability, such as stem-and-leaf plots, box plots, and histograms.

In this chapter, the mean $\bar{x}$ we defined previously, will now be referred as **sample mean**.

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j = \frac{1}{n} (x_1 + x_2 + \cdots + x_n) .$$

We call n the **sample size**, and similar to mean, the variance s^2 is called the **sample variance**:

$$s^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2 = \frac{1}{n-1} [(x_1 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2] ,$$

and its positive square root, s is the **sample standard deviation**.

$\bar{x}$, s^2 , s are called **sample parameters** of a dataset.

Information: Optimal Read: John Snow and the Founding of Epidemiology

John Snow (1813 - 1858) was an English physician and a leader in the development of anaesthesia and medical hygiene. He is considered one of the founders of modern epidemiology and early germ theory, in part because of his work in tracing the source of a cholera outbreak in London's Soho, which he identified as a particular public water pump. Snow was a skeptic of the then-dominant miasma theory stated diseases such as cholera and bubonic plague were caused by pollution or a noxious form of "bad air". The germ theory of disease had not yet been developed, so Snow did not understand the mechanism by which the disease was transmitted. His observation of the evidence led him to discount the theory of foul air. He first published his theory in an 1849 essay, *On the Mode of Communication of Cholera*, followed by a more detailed treatise in 1855 incorporating the results of his investigation of the role of the water supply in the Soho epidemic of 1854.

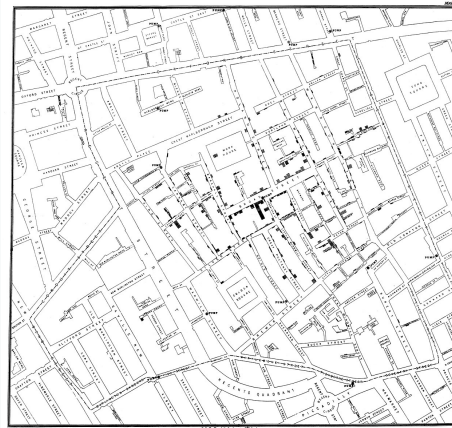


Figure 2.1: Original map by John Snow showing the clusters of cholera cases in the London epidemic of 1854, drawn and lithographed by Charles Cheffins.

By talking to local residents, he identified the source of the outbreak as the public water pump on Broad Street (now Broadwick Street). Although Snow's chemical and microscope examination of a water sample from the Broad Street pump did not conclusively prove its danger, his studies of the pattern of the disease were convincing enough to persuade the local council to disable the well pump by removing its handle.

This action has been commonly credited as ending the outbreak, but Snow observed that the epidemic may have already been in rapid decline.

2.2 Point Estimation of Parameters

Before we dive deep into statistics, let's spend some time to learn the most basic practical tasks in statistics and corresponding statistical methods to accomplish them. The first is point **estimation of parameters**, that is, of **quantities** appearing in distributions:

such as p in the binomial distribution and μ and σ in the normal distribution.

A **point estimate** of a parameter is a number,¹⁰ which is computed from a given sample and serves as an **approximation of the unknown exact value** of the parameter of the population. An interval estimate is an interval¹¹ obtained from a sample. Think of it as a value which is a sensible guess for that parameter.

¹⁰which is a point on the real line.

¹¹also known as confidence interval.

Estimation of parameters is of great practical importance in many applications.

As an approximation¹² of the mean of a population we may take the mean $\bar{x}$ of a corresponding sample. This gives the estimate $\hat{\mu} = \bar{x}$ for μ , that is,

$$\hat{\mu} = \bar{x} = \frac{1}{n} (x_1 + \cdots + x_n), \quad (2.1)$$

where n is the sample size. Similarly, an estimate $\hat{\sigma}^2$ for the variance of a population is the variance s^2 of a corresponding sample, that is:

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2. \quad (2.2)$$

As can be seen, both Eq. (2.1) and Eq. (2.2) are estimates¹³ of parameters for distributions in which μ or σ^2 appear explicitly as parameters, such as the normal and Poisson distributions.

¹²To describe something which is an approximation or an educated guess, we use hat (i.e., $\hat{x}$) notation. This is applicable for fields in statistics, machine learning or data science.

An estimator is not expected to estimate the population parameter without error. We do not expect $\bar{x}$ to estimate μ exactly, but we certainly hope that it is not far off.

¹³There are numerous practical examples for parameter estimation, such as using political polls to estimate voter turnout or the proportion of voters who will vote for a particular candidate, estimating the proportion of consumers interested in a product before massive production, or estimating the success rate of a product

For the binomial distribution, $p = \mu/n$. From Eq. (2.1) we obtain for p the estimate:

$$\hat{p} = \frac{\bar{x}}{n}.$$

It is important to mention Eq. (2.1) is a special case of the so-called **method of moments**. Here, the parameters to be estimated are expressed in terms of the moments of the distribution. In the resulting formulas, those moments of the distribution are replaced by the corresponding moments of the sample, which gives the estimates. Here the k^{th} moment of a sample $x_1, \dots, x_n$ is:

$$m_k = \frac{1}{n} \sum_{j=1}^n x_j^k.$$

2.2.1 Maximum Likelihood Method

Another method for obtaining estimates is the so-called **maximum likelihood method** conceived by R. A. Fisher.¹⁴ To explain it, we consider a discrete (or continuous) random variable X whose probability function (or density) $f(x)$ depends on a single parameter θ . We take a corresponding sample of n **independent** values $x_1, \dots, x_n$. Then in the discrete case the probability given a sample of size n consists precisely of those n values is

$$L(x_1, x_2; \theta_1) = f(x_1) f(x_2) \cdots f(x_n). \quad (2.3)$$

In the continuous case the probability that the sample consists of values in the small intervals $x_j \leq x \leq x_j + \Delta x$ where $(j = 1, 2, \dots, n)$ is

$$f(x_1) \Delta x f(x_2) \Delta x \cdots f(x_n) \Delta x = L(\Delta x)^n, \quad (2.4)$$

as $f(x_j)$ depends on θ , the function L in Eq. (2.4) given by Eq. (2.3) depends on $x_1, \dots, x_n$ and θ .

We imagine $x_1, \dots, x_n$ to be given and **fixed**.

Then L is a function of θ , which is called the **likelihood function**. The basic idea of the maximum likelihood method is quite simple, as follows.

We choose an approximation for the unknown value of θ for which L is **as large as possible**.

If L is a differentiable function of θ , a necessary condition for L to have a maximum in an interval¹⁵ is

$$\frac{\partial L}{\partial \theta} = 0 \quad (2.5)$$

A solution of Eq. (2.5) depending on $x_1, \dots, x_n$ is called a **maximum likelihood estimate** for θ . We may replace Eq. (2.5) by:¹⁶

$$\frac{\partial \ln L}{\partial \theta} = 0 \quad (2.6)$$

as $f(x_j) > 0$, a maximum of L is in general positive, and $\ln L$ is a monotone increasing function of L . This often simplifies calculations due to constant multiplication of the same function n times to get L . The use of $\ln$ turns the exponentiation parameter to a multiplication parameter.

Several Parameters

If the distribution of X involves r parameters $\theta_1, \dots, \theta_r$, then instead of Eq. (2.5) we have the r conditions $\partial \ln L / \partial \theta_1, \dots, \partial \ln L / \partial \theta_r = 0$, and instead of Eq. (2.6) we have:

$$\frac{\partial \ln L}{\partial \theta_1} = 0, \dots, \frac{\partial \ln L}{\partial \theta_r} = 0.$$



¹⁴Considered the father of modern statistics. For his work in statistics, he has been described as "a genius who almost single-handedly created the foundations for modern statistical science" and "the single most important figure in 20th century statistics". Fisher has also been praised as a pioneer of the Information Age. His work on a mathematical theory of information ran parallel to the work of Claude Shannon and Norbert Wiener, though based on statistical theory.

¹⁵not at the boundary.

¹⁶It is worth noting that if the function is **NOT** derivable in some points of the domain, critical point is not a maximum, and the global maximum occurs on the boundary of the domain, it may be required to take the 2nd derivative [1]. However, if you know that the function you are trying to estimate the parameter has one (1) critical point, a single derivative would suffice.

Exercise 2.1 Maximum Likelihood of Gaussian Distribution

Find maximum likelihood estimates for $\theta_1 = \mu$ and $\theta_2 = \sigma$ in the case of the normal distribution. **SOLUTION**

We obtain the likelihood function:

$$L = \left(\frac{1}{\sqrt{2\pi}} \right)^n \left(\frac{1}{\sigma} \right)^n e^{-h}$$

$$\text{where } h = \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2.$$

Taking logarithms, we have

$$\ln L = -n \ln \sqrt{2\pi} - n \ln \sigma - h.$$

The first equation for the parameters is $\frac{\partial \ln L}{\partial \mu} = 0$, written out:

$$\frac{\partial \ln L}{\partial \mu} = -\frac{\partial h}{\partial \mu} = \frac{1}{\sigma^2} \sum_{j=1}^n (x_j - \mu) = 0,$$

$$\text{therefore } \sum_{j=1}^n x_j - n\mu = 0.$$

The solution is the desired estimate $\hat{\mu}$ for μ : we find

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n x_j = \bar{x}.$$

The second equation for the parameter is $\partial \ln L / \partial \sigma = 0$, written out

$$\frac{\partial \ln L}{\partial \sigma} = -\frac{n}{\sigma} - \frac{\partial h}{\partial \sigma} = -\frac{1}{\sigma} + \frac{1}{\sigma^3} \sum_{j=1}^n (x_j - \mu)^2 = 0.$$

Replacing μ by $\hat{\mu}$ and solving for σ^2 , we obtain the estimate:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})^2 \quad \blacksquare$$

Exercise 2.2 Maximum Likelihood of Poisson Distribution

Consider a Poisson distribution with probability mass function:

$$f(x|\mu) = \frac{e^{-\mu} \mu^x}{x!} \quad \text{where } x = 0, 1, 2, \dots$$

Suppose that a random sample $x_1, x_2, \dots, x_n$ is taken from the distribution. What is the maximum likelihood estimate of μ ? **SOLUTION** The likelihood function is

$$L(x_1, x_2, \dots, x_n; \mu) = \prod_{i=1}^n f(x_i|\mu) = \frac{e^{-n\mu} \sum_{i=1}^n x_i}{\prod_{i=1}^n x_i!}.$$

Now consider its logarithmic representation:

$$\ln L(x_1, x_2, \dots, x_n; \mu) = -n\mu + \sum_{i=1}^n x_i \ln \mu - \ln \prod_{i=1}^n x_i!$$

And taking its partial derivative to the parameter gives:

$$\frac{\partial \ln L(x_1, x_2, \dots, x_n; \mu)}{\partial \mu} = -n + \sum_{i=1}^n \frac{x_i}{\mu}.$$

Solving for $\hat{\mu}$, the maximum likelihood estimator, involves setting the derivative to zero and solving for the parameter. Therefore,

$$\hat{\mu} = \sum_{i=1}^n \frac{x_i}{n} = \bar{x}$$

If you were to test it, the second derivative of the log-likelihood function is **negative**, which implies that the solution above indeed is a maximum. As μ is the mean of the Poisson distribution, the sample average would certainly seem like a reasonable estimator $\blacksquare$.

Applications

There are numerous applications for **Maximum Likelihood Estimation**, for example when dealing with design and control systems in engineering [2], it is used to estimate system parameters based on noisy measurements. In the financial sector [3], it can help estimate parameters of models like Black-Scholes-Merton [4], which describe the dynamics of financial derivatives, and in Biology, it is used in genetic mapping [5] and genome-wide association studies [6].

Exercise 2.3 For Science

Suppose ten (10) rats are used in a biomedical study where they are injected with cancer cells and then given a cancer drug that is designed to increase their survival rate. The survival times, in months, are:

14 17 27 18 12 8 22 13 19 12

Assume exponential distribution applies which is given as:

$$f(x, \beta) = \begin{cases} \frac{1}{\beta} \exp \frac{x}{\beta}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Give a maximum likelihood estimate of the mean survival time.

SOLUTION

We know that the probability density function for the

exponential random variable X . Therefore, the log-likelihood function for the data, given $n = 10$, is:

$$\ln L(x_1, x_2, \dots, x_{10}; \beta) = -10 \ln \beta - \frac{1}{\beta} \sum_{i=1}^{10} x_i.$$

Setting

$$\frac{\partial \ln L}{\partial \beta} = -\frac{10}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^{10} x_i = 0 \quad \text{which means}$$

$$\hat{\beta} = \frac{1}{10} \sum_{i=1}^{10} x_i = \bar{x} = 16.2 \quad \blacksquare$$

Evaluating the second derivative of the log-likelihood function at the value $\hat{\beta}$ above gives a negative value. As a result, the estimator of the parameter β , the population mean, is the sample average $\bar{x}$.

Exercise 2.4 Sampling the Population

It is known that a sample consisting of the values:

12 11.2 13.5 12.3 13.8 11.9

comes from a population with the density function:

$$f(x; \theta) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x > 1, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\theta > 0$. Find the maximum likelihood estimate of θ .

SOLUTION

The likelihood function of n observations from this population can be written as:

$$L(x_1, x_2, \dots, x_{10}; \theta) = \prod_{i=1}^n \frac{\theta}{x_i^{\theta+1}} = \frac{\theta^n}{\left(\prod_{i=1}^n x_i\right)^{\theta+1}}.$$

which implies that

$$\ln L(x_1, x_2, \dots, x_{10}; \theta) = n \ln \theta - (\theta + 1) \sum_{i=1}^n \ln x_i.$$

Setting $0 = \partial \ln L / \partial \theta = n/\theta - \sum_{i=1}^n \ln(x_i)$ results in

$$\hat{\theta} = \frac{n}{\sum_{i=1}^n \ln x_i} = 0.3970 \quad \blacksquare.$$

Since the second derivative of L is $-n/\theta^2$, which is always negative, the likelihood function does achieve its maximum value at $\hat{\theta}$.

Information: Exponential Distribution

It is the probability distribution of the distance between events in a Poisson point process, i.e., a process in which events occur continuously and independently at a constant average rate, such as time between production errors, or length along a roll of fabric in the weaving manufacturing process. It also has the feature of being memoryless:¹⁷

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Here $\lambda > 0$ is the parameter of the distribution, often called the **rate parameter**. The exponential distribution occurs naturally when describing the lengths of the inter-arrival times in a homogeneous Poisson process. Such as the time until a radioactive particle decays, or the time between clicks of a Geiger counter, or the time between receiving one telephone call and the next.

¹⁷i.e., previous failures or elapsed time does not affect future trials.

2.3 Confidence Intervals

Confidence intervals¹⁸ for an unknown parameter θ of some distribution (e.g., $\theta = \mu$) are intervals $\theta_1 \leq \theta \leq \theta_2$ which contain θ , **NOT** with certainty but with a **high probability** γ , which we can choose where 95% and 99% are popular choices. Such an interval is calculated from a sample. $\gamma = 95\%$ means probability $1 - \gamma = 5\% = 1/20$ of being wrong. Instead of writing $\theta_1 \leq \theta \leq \theta_2$, we denote this more **distinctly** by writing:

$$\text{CONF}_\gamma \{ \theta_1 \leq \theta \leq \theta_2 \} \quad (2.7)$$

Such a special symbol, CONF, seems worthwhile to avoid the misunderstanding that θ **must** lie between θ_1 and θ_2 .

γ is called the **confidence level**, and θ_1 and θ_2 are called the **lower** and **upper confidence limits**, respectively and **depend** on the γ value. The larger we **choose** γ , the smaller is the error probability $1 - \gamma$, but the longer is the confidence interval.

As $\gamma \rightarrow 1$, the interval goes to infinity.

The choice of γ depends on the type of application.

In taking no umbrella, a 5% chance of getting wet is **NOT** a problem. In a medical decision of life or death, a 5% chance of being wrong may be too large and a 1% chance of being wrong ($\gamma = 99\%$) may be more desirable.

Confidence intervals are more valuable than point estimates. We can take the midpoint of Eq. (2.7) as an approximation of θ and half the length of Eq. (2.7) as an error bound.¹⁹

θ_1 and θ_2 in Eq. (2.7) are calculated from a sample $x_1, \dots, x_n$. These are n observations of a random variable X . Now comes a **trick**.

We regard $x_1, \dots, x_n$ as single observations of n random variables $x_1, \dots, x_n$.²⁰ Then $\theta_1 = \theta_1(x_1, \dots, x_n)$ and $\theta_2 = \theta_2(x_1, \dots, x_n)$ in Eq. (2.7) are observed values of two (2) random variables $\Theta_1 = \Theta_1(x_1, \dots, x_n)$ and $\Theta_2 = \Theta_2(x_1, \dots, x_n)$. The condition Eq. (2.7) involving γ can now be written:²¹

$$P(\Theta_1 \leq \theta \leq \Theta_2) = \gamma.$$

Let us see what all this means in concrete practical cases.

In each case in this section we shall first state the steps of obtaining a confidence interval in the



¹⁸Established by Jerzy Neyman. He proposed and studied randomised experiments in 1923. Furthermore, his paper *On the Two Different Aspects of the Representative Method: The Method of Stratified Sampling and the Method of Purposive Selection* [7], given at the Royal Statistical Society in 1934, was the groundbreaking event leading to modern scientific sampling. He introduced the confidence interval in his paper in 1937 [8]. Another noted contribution is the Neyman-Pearson lemma, the basis of hypothesis testing [9].

¹⁹not in the strict sense of numerical means, but except for an error whose probability we know.

²⁰with the same distribution as X

²¹As an example, to say that there is 95% confidence is shorthand for "95% of all possible samples of a given size from this population will result in an interval that captures the unknown parameter."

form of a table, then consider a typical example, and finally justify those steps theoretically.

For Mean with Known Variance in Normal Distribution

The method of tackling this problem is as follows:

1. Choose a confidence level for γ .²²
2. Determine the corresponding c :

γ	0.90	0.95	0.99	0.999
c	1.645	1.960	2.576	3.291

Table 2.1: Useful c values based on a given confidence (γ) value.

²²95%, 99%, depending on the application.

3. Calculate the mean $\bar{x}$ of the sample $x_1, \dots, x_n$.
4. Calculate $k = c\sigma/\sqrt{n}$. The confidence interval for μ is

$$\text{CONF}_{\gamma} \{ \bar{x} - k \leq \mu \leq \bar{x} + k \}.$$

Exercise 2.5 Confidence Interval for Mean with known Variance in Normal Distribution

Determine 95% confidence interval for the mean of a normal distribution with variance $\sigma^2 = 9$, using a sample of $n = 100$ values with mean $\bar{x} = 5$. **SOLUTION**

1. First we define γ as 0.95 based on the 95% confidence.
2. Then looking at our reference table find the corresponding c which equals 1.960.
3. $\bar{x} = 5$ is given.

4. We need:

$$k = c \frac{\sigma}{\sqrt{n}} = 1.960 \frac{3}{\sqrt{100}} = 0.588$$

Therefore

$$\bar{x} - k = 4.412 \quad \text{and} \quad \bar{x} + k = 5.588$$

and the confidence interval is:

$$\text{CONF}_{0.95} \{ 4.412 \leq \mu \leq 5.588 \} \quad \blacksquare$$

Theory 2.19: Sum of Independent Normal Random Variables

Let $x_1, \dots, x_n$ be independent normal random variables each of which has mean μ and variance σ^2 .

Then the following holds:

- a. The sum $X_1 + \dots + X_n$ is normal with mean $n\mu$ and variance $n\sigma^2$.
- b. The following random variable $\bar{X}$ is normal with mean μ and variance σ^2/n .

$$\bar{X} = \frac{1}{n} (X_1 + \dots + X_n) \quad (2.8)$$

c. The following random variable Z is normal with mean 0 and variance 1.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Exercise 2.6 Sample Size Needed for a Confidence Interval of Prescribed Length

How large must n be in the Example **Confidence Interval for mean with known variance in Normal Distribution** to obtain a 95% confidence interval of length $L = 0.4$?

SOLUTION

The interval in Example **Confidence Interval for mean with known variance in Normal Distribution** has the length:

$$L = 2k = 2c\sigma/\sqrt{n}.$$

Solving for n , we obtain

$$n = \left(\frac{2c\sigma}{L} \right)^2$$

In the present case the answer is:

$$n = \left(\frac{2 \times 1.96 \times 3}{0.4} \right)^2 \approx 870 \quad \blacksquare$$

For Mean of the Normal Distribution with Unknown Variance

For practical applications, σ^2 is frequently **unknown**. Then the method described previously does **NOT** help and the whole theory changes, although the steps of determining a confidence interval for μ remain quite similar.

We see that k differs from previous method, namely, the sample standard deviation s has taken the place of the unknown standard deviation σ of the population as it is now the variance we are trying to estimate, and c now depends on the sample size n and must be determined from **Tbl. ??** to **Tbl. ??**. That table lists values z for given values of the distribution function.

$$F(z) = K_m \int_{-\infty}^x \left(1 + \frac{u^2}{m} \right)^{-(m+1)/2} du$$

of the t -distribution. Here, $m = 1, 2, \dots$ is a parameter, called the **number of degrees of freedom** of the distribution.²³ In the present case, $m = n - 1$ where n is the number of sample we have to determine variance. The constant K_m is such that $F(\infty) = 1$. By integration it turns out that:²⁴

$$K_m = \frac{\Gamma\left(\frac{1}{2}m + \frac{1}{2}\right)}{\sqrt{m\pi}\Gamma\left(\frac{1}{2}m\right)},$$

where Γ is the gamma function.²⁵

The method of tackling is this problem is as follows:

1. Choose a confidence level γ .²⁶
2. Determine the solution c of the equation,

$$F(c) = \frac{1}{2}(1 + \gamma)$$

from the table of the t -distribution with $m = n - 1$ degrees of freedom

3. Compute the mean $\bar{x}$ and the variance s^2 of the sample $x_1, \dots, x_n$.

²³abbreviated d.f.

²⁴For most practical application a reference Table would suffice to determine the parameter.

²⁵Do not worry if these equations do not make sense as it is here for literature purposes.

²⁶95%, 99%, or the like.

4. Compute $k = cs/\sqrt{n}$. The confidence interval is:

$$\text{CONF}_\gamma \{\bar{x} - k \leq \mu \leq \bar{x} + k\}.$$

This illustrates that 2.1²⁷ provides shorter confidence intervals than Table XX, which enforces the idea of getting a shorter interval range by increasing the sample size.

Exercise 2.7 Confidence Interval for Mean of Normal Distribution with Unknown Variance

The five (5) independent measurements of flash point of Diesel oil (D-2) gave the values (in °F):

144 147 146 142 144

If we assume normality, determine a 99% confidence interval for the mean. **SOLUTION**

1. $\gamma = 0.99$ is required based on 99% confidence level.
2. $F(c) = \frac{1}{2}(1 + \gamma) = 0.99$ and looking at the reference table with $n - 1 = 4$ d.f., which gives $c = 4.60$.
3. Calculating the mean and the variance gives $\bar{x} = 144.6$ and $s = 3.8$,

4. $k = \sqrt{3.8} \times 4.60/\sqrt{5} = 4.01$. Therefore the confidence interval is:

$$\text{CONF}_{0.99} \{140.5 \leq \mu \leq 148.7\} \blacksquare$$

If the variance σ^2 were known and equal to the sample variance s^2 , therefore $\sigma^2 = 3.8$, then the Reference Table would give:

$$k = \frac{c\sigma}{\sqrt{n}} = 2.576 \frac{\sqrt{3.8}}{\sqrt{5}} = 2.25$$

and

$$\text{CONF}_{0.99} \{142.35 \leq \mu \leq 146.85\} \blacksquare$$

We see that the present interval is almost twice as long as that with a known variance $\sigma^2 = 3.8$.

Theory 2.20: Student's t-Distribution

Let $x_1, \dots, x_n$ be independent normal random variables with the same mean μ and the same variance σ^2 . Then the random variable:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t-distribution²⁸ with $n - 1$ degrees of freedom (d.f.). Here $x_1, \dots, x_n$ is given by Eq. (2.8) and S is defined as:

$$S^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{X})^2. \quad (2.9)$$

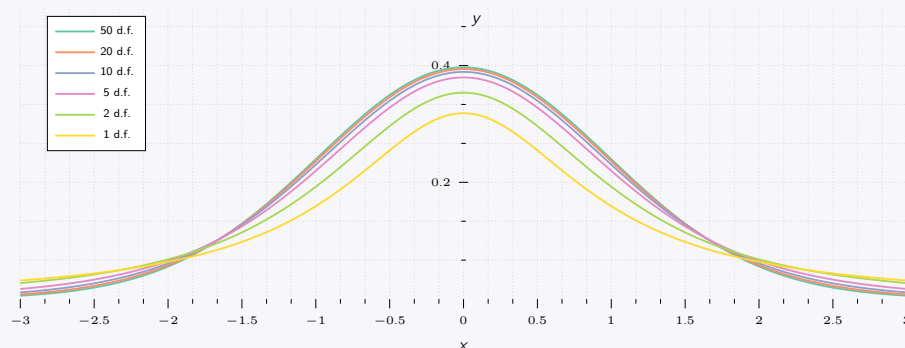


Figure 2.2: The student-t distribution with different degrees of freedom m .



²⁸William Gosset (1876 - 1937) was an English statistician, chemist and brewer who served as Head Brewer of Guinness and Head Experimental Brewer of Guinness and was a pioneer of modern statistics. He published his results under the pen name student.

Estimating the Variance of the Normal Distribution

The method for calculating the confidence interval is similar to the previous methods, with slight change in some steps which are as follows:

1. Choose a confidence level γ .²⁹
2. Determine solutions c_1 and c_2 of the equations:

$$F(c_1) = \frac{1}{2}(1 - \gamma) \quad \text{and} \quad F(c_2) = \frac{1}{2}(1 + \gamma).$$

where the necessary values are calculated from the table of the chi-square distribution with $n - 1$ degrees of freedom., given in **Tbl. ??** to **Tbl. ??**.

3. Calculate $(n - 1)s^2$, where s^2 is the variance of the sample $x_1, \dots, x_n$.
4. Calculate $k_1 = (n - 1)s^2/c_1$ and $k_2 = (n - 1)s^2/c_2$. The confidence interval is

$$\text{CONF}_\gamma\{k_2 \leq \sigma^2 \leq k_1\}.$$

²⁹as usual this can be 95%, 99%, or the like.

Theory 2.21: Chi-Square Distribution

Under the assumptions in Theorem **Student's t-Distribution** the random variable:

$$Y = (n - 1) \frac{S^2}{\sigma^2}$$

with S_2 given by Eq. (2.9) has a chi-square distribution with $n - 1$ degrees of freedom.

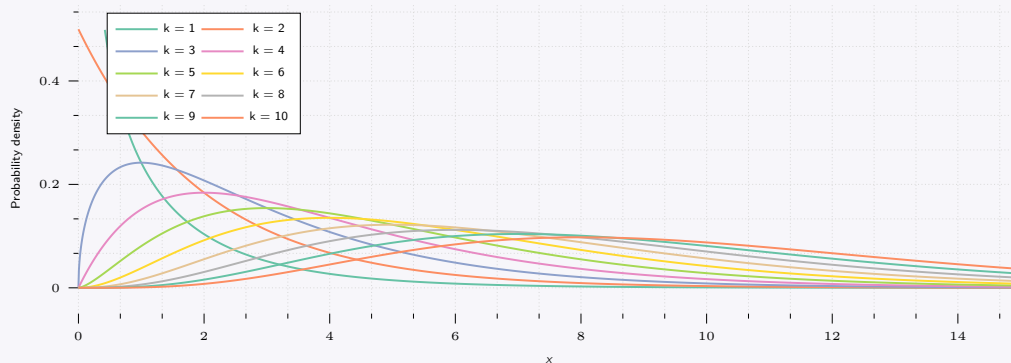


Figure 2.3: Chi-square distribution with different degrees of freedom.

The chi-squared distribution, which can be seen in **Fig. 2.3** is used primarily in **hypothesis testing**, and to a lesser extent for confidence intervals for population variance when the underlying distribution is **normal**. Unlike more widely known distributions such as the normal distribution and the exponential distribution, the chi-squared distribution is not as often applied in the direct modelling of natural phenomena.

The primary reason for which the chi-squared distribution is extensively used in hypothesis testing is

its relationship to the normal distribution. Many hypothesis tests use a test statistic, such as the **t-student**. For these hypothesis tests, as the sample size n increases, the sampling distribution of the test statistic approaches the normal distribution.³⁰ Because the test statistic (t) is asymptotically normally distributed, provided the sample size is sufficiently large, the distribution used for hypothesis testing may be approximated by a normal distribution.

So wherever a normal distribution could be used for a hypothesis test, a chi-squared distribution could be used.

³⁰This is the result of the central limit theorem.

2.4 Testing of Hypotheses and Making Decisions

The ideas of confidence intervals and of tests³¹ are the two (2) most important ideas in modern statistics. In a statistical test we make inference from sample to population through testing a **hypothesis**, resulting from experience or observations, from a theory or a quality requirement, and so on.

In many cases the result of a test is used as a basis for a **decision**:

to buy, or not to buy a certain model of car, depending on a test of the fuel efficiency ($\text{km} \cdot \text{L}^{-1}$), or, to apply some medication, depending on a test of its effect; to proceed with a marketing strategy, depending on a test of consumer reactions, etc.

As with most abstract mathematical concepts, it is better to explain such a test in terms of a typical example and then introduce the corresponding standard notions of statistical testing.

Example 2.8: Test of a Hypothesis

Let's say we want to buy 100 coils of a certain kind of wire, provided we can verify the manufacturer's claim, the wire has a specific strength of $\mu = \mu_0 = 200 \text{ kN} \cdot \text{m} \cdot \text{kg}^{-1}$, or more.

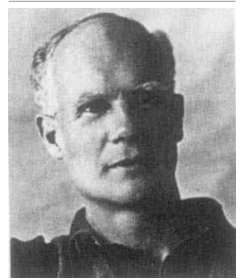
This is a test of the hypothesis:³² $\mu = \mu_0 = 200$. We shall **NOT** buy the wire if the statistical tests shows that actually $\mu = \mu_1 < \mu_0$, the wire is weaker, the claim does **NOT** hold. μ_1 is called the **alternative** of the test.³³ We shall **accept** the hypothesis if the test suggests that it is true, except for a small error probability α , called the **significance level** of the test.

Otherwise we reject the hypothesis.

Hence α is the probability of rejecting a hypothesis although it is true. The choice of α is up to us, 5% and 1% are popular values.

For the test we need a sample. We randomly select 25 coils of the wire, cut a piece from each coil, and determine the breaking limit experimentally. Suppose that this sample of $n = 25$ values of the breaking limit has the mean $\bar{x} = 197 \text{ kN} \cdot \text{m} \cdot \text{kg}^{-1}$, which is somewhat less than the claim, and the standard deviation $s = 6 \text{ kN} \cdot \text{m} \cdot \text{kg}^{-1}$.

At this point we could only speculate when this difference $197 - 200 = -3$ is due to randomness, is a chance effect, or whether it is **significant**, due to the actual inferior quality of the wire. To continue beyond speculation requires probability theory, as follows.



³¹The modern development of tests are generally attributed to Egon Sharpe Pearson and Neyman whom was mentioned previously. Egon Sharpe was one of three children of Karl Pearson and Maria, and, like his father, a British statistician. He is known throughout the world as co-author of the Neyman-Pearson theory of testing statistical hypotheses, and responsible for many important contributions to problems of statistical inference and methodology, especially in the development and use of the likelihood ratio criterion.

³²also called null hypothesis.

³³or alternative hypothesis

We assume that the blocking limit is normally distributed. Then

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

with $\mu = \mu_0$ has a **t-distribution** with $n - 1$ degrees of freedom ($n - 1 = 24$ for our sample). Also $\bar{x} = 197$ and $s = 6$ are observed values of $\bar{X}$ and S to be used later. We can now choose a significance level, say, $\alpha = 5\%$. From the Reference Table, we then obtain a critical value c such that $P(T \leq c) = \alpha = 5\%$. For $P(T \leq \bar{c}) = 1 - \alpha = 95\%$ the table gives $\bar{c} = 1.71$, so that $c = -\bar{c} = -1.71$ because of the symmetry of the distribution shown in **Fig. 2.4**.

We now reason as follows—this is the crucial idea of the test. If the hypothesis is true, we have a chance of only α ($= 5\%$) that we observe a value t of T (calculated from a sample) that will fall between $-\infty$ and -1.71 . Hence, if we nevertheless do observe such a t , we start that the hypothesis cannot be true and we reject it.

A simple calculation gives:

$$T = \frac{(107 - 200)}{6/\sqrt{25}} = -2.5,$$

as an observed value of T . Since $-2.5 < -1.71$, we reject the hypothesis, the manufacturer's claim, and accept the alternative result of $\mu = \mu_4 < 200$, which means the wire seems to be weaker than claimed ■

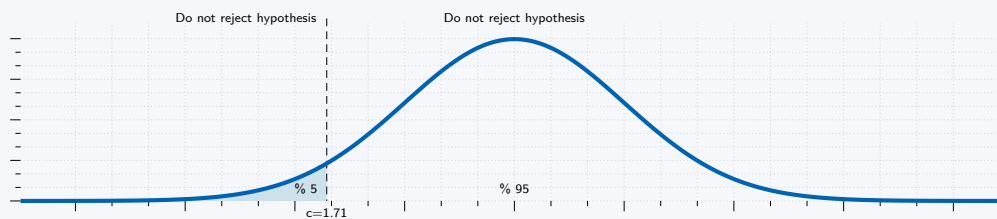


Figure 2.4: The t -distribution used in example. As can be seen, anything left of the critical line would tell us to reject the hypothesis, whereas if the t value lies on the RHS, then the test would tell us the null hypothesis is true.

This aforementioned example perfectly captures the **steps of a test**:

1. Formulate the **hypothesis** $\theta = \theta_0$ to be tested. In our previous example it is $\theta_0 = \mu_0$.
2. Formulate an **alternative** $\theta = \theta_1$, which in our example is $\theta_1 = \mu_1$.
3. Choose a **significance level** α with values such as 5%, 1%, or, 0.1%.
4. Use a random variable $\hat{\Theta} = g(x_1, \dots, x_n)$ whose distribution depends on the hypothesis and on the alternative, and this distribution is known in both cases.

Determine a critical value c from the distribution of $\hat{\Theta}$, assuming the hypothesis to be true. In the example, $\hat{\Theta} = T$, and c is, obtained from $P(T \leq c) = \alpha$.

5. Use a sample $x_1, \dots, x_n$ to determine an observed value $\hat{\theta} = g(x_1, \dots, x_n)$ of $\hat{\Theta}$, where in our example it is t .
6. Accept or reject the hypothesis, depending on the size of $\hat{\theta}$ **relative** to c .

There are two (2) important facts require further discussion and careful attention.

1. The choice of an alternative. In the example, $\mu_1 < \mu_0$, but other applications may require $\mu_1 > \mu_0$ or $\mu_1 \neq \mu_0$ as not all application can fit the same criterion. Some applications may require an upper-bound, some lower-bound.
2. Addressing errors. We know that α , the significance level of the test, is the probability of reflecting a **true** hypothesis. And we shall discuss the probability β of accepting a **false** hypothesis.

One-Sided and Two-Sided Alternatives

Let θ be an **unknown parameter** in a distribution, and suppose we want to test the hypothesis $\theta = \theta_0$.

Then there are three (3) main kinds of alternatives, namely,

$$\theta > \theta_0 \quad (2.10)$$

$$\theta < \theta_0 \quad (2.11)$$

$$\theta \neq \theta_0 \quad (2.12)$$

Here Eq. (2.10), and Eq. (2.11) are **one-sided alternatives**, and Eq. (2.12) is a **two-sided alternative**.

We call rejection region³⁴ the region such that we reject the hypothesis if the observed value in the test falls in this region. In Eq. (2.10) the critical c lies to the right of θ_0 because so does the alternative. Hence the rejection region extends to the right. This is called a **right-sided test**. In Eq. (2.11) the critical c lies to the left of θ_0 (as in **Test of a Hypothesis**), the rejection region extends to the left, and we have a **left-sided test**. These are one-sided tests. In Eq. (2.12)

³⁴or called the critical region

All three (3) kinds of alternatives occur in practical problems. For example, Eq. (2.10) may arise if θ_0 is the maximum tolerable inaccuracy of a voltmeter or some other instrument. Alternative Eq. (2.11) may occur in testing strength of material, as in **Test of a Hypothesis**. Finally, θ_0 in Eq. (2.12) may be the diameter of axle-shafts, and shafts that are too thin or too thick are equally undesirable, so that we have to watch for deviations in both directions.

2.4.1 Errors in Tests

Tests always involve **risks of making false decisions**:

I Rejecting a true hypothesis (Type I error)

■ α = Probability of making a Type I error.

II Accepting a false hypothesis (Type II error).

■ β = Probability of making a Type II error.

Clearly, we cannot avoid these errors.

No absolutely certain conclusions about populations can be drawn from samples.

But we show there are ways and means of choosing suitable levels of risks, that is, of values α and β . The choice of α depends on the nature of the problem.³⁵

³⁵e.g., a small risk $\alpha = \%1$ is used if it is a matter of life or death.

Let us discuss this systematically for a test of a hypothesis $\theta = \theta_0$ against an alternative that is a single number θ_1 , for simplicity. We let $\theta_1 > \theta_0$, so that we have a **right-sided test**. For a left-sided

or a two-sided test the discussion is quite similar.

We choose a critical $c > \theta_0$. From a given sample $x_1, \dots, x_n$ we then compute a value:

$$\hat{\theta} = g(x_1, \dots, x_n)$$

with a suitable g .

the choice of g will be a main point of our further discussion; for instance, take $g = (x_1 + \dots + x_n) / n$ in the case in which θ is the mean.

If $\hat{\theta} > c$, we **reject the hypothesis**. If $\hat{\theta} \leq c$, we accept it. Here, the value $\hat{\theta}$ can be regarded as an observed value of the random variable

$$\hat{\Theta} = g(x_1, \dots, x_n)$$

because x_j may be regarded as an observed value of X_j where $j = 1, \dots, n$. In this test there are two (2) possibilities of making an error, as follows.

Type I Error The hypothesis is true but is rejected³⁶ because Θ assumes a value $\hat{\theta} > c$. Obviously, the probability of making such an error equals

³⁶hence the alternative is accepted.

$$P(\hat{\Theta} > c)_{\theta} = \theta_0 = \alpha. \quad (2.13)$$

α is called the **significance level** of the test, as mentioned before.

Type II Error The hypothesis is false but is accepted because $\hat{\Theta}$ assumes a value $\hat{\theta} \leq c$. The probability of making such an error is denoted by β ; Therefore:

$$P(\hat{\Theta} \leq c)_{\theta} = \theta_0 = \beta. \quad (2.14)$$

$\eta = 1 - \beta$ is called the **power** of the test. Obviously, the power η is the probability of avoiding a Type II error.

Formulas Eq. (2.13) and Eq. (2.14) show that both α and β depend on c , and we would like to choose c so these probabilities of making errors are as small as possible. But the important **Fig. 2.5** shows that these are conflicting requirements because to let α decrease we must shift c to the right, but then β increases. In practice we first choose α (5%, sometimes 1%), then determine c , and finally compute β . If β is large so that the power $\eta = 1 - \beta$ is small, we should repeat the test, choosing a larger sample, for reasons that will appear shortly. If the alternative is **NOT** a single number but is of the form Eq. (2.10)-Eq. (2.12), then β becomes a function of θ . This function $\beta(\theta)$ is called the operating characteristic (OC) of the test and its curve the OC curve. Clearly, in this case $\eta = 1 - \beta$ also depends on θ . This function $\eta(\theta)$ is called the **power function** of the test.

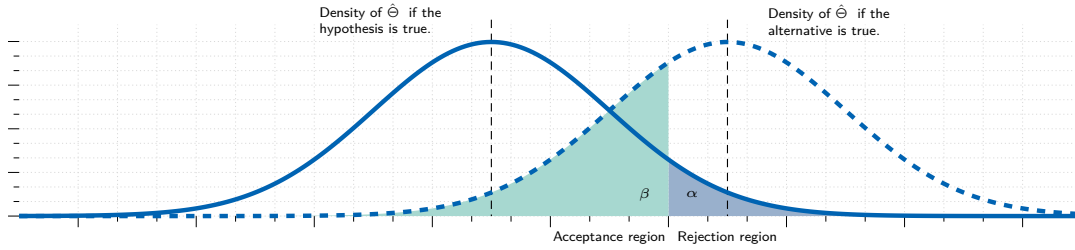


Figure 2.5: Illustration of Type I and II errors in testing a hypothesis $\theta = \theta_0$ against an alternative $\theta = \theta_0$.

Of course, from a test that leads to the acceptance of a certain hypothesis θ_0 , it does **NOT** follow that this is the only possible hypothesis or the best possible hypothesis. Hence the terms “not reject” or “fail to reject” are perhaps better than the term “accept”.

Exercise 2.9 Test for the Mean of the Normal Distribution with Known Variance

Let X be a normal random variable with variance $\sigma^2 = 9$. Using a sample of size $n = 10$ with mean $\bar{x}$, test the hypothesis $\mu = \mu_0 = 24$ against the three (3) kinds of alternatives, namely,

- (a) $\mu > \mu_0$ (b) $\mu < \mu_0$ (c) $\mu \neq \mu_0$

SOLUTION We choose the significance level $\alpha = 0.05$ as it is customary at this point. An estimate of the mean will be obtained from:

$$\bar{X} = \frac{1}{n} (X_1 + \cdots + X_n).$$

If the hypothesis is true, $\bar{X}$ is normal with mean $\mu = 24$ and variance $\sigma^2/n = 0.9$. Therefore we may obtain the critical value c from X .

Right-Sided Test We determine c from

$$P(\bar{X} > c)_{\mu=24} = \alpha = 0.05$$

that is,

$$P(\hat{X} \leq c)_{\mu=24} = \Phi\left(\frac{c-24}{\sqrt{0.9}}\right) = 1 - \alpha = 0.95.$$

Reverse engineering Tbl. ?? by looking for 0.95 percentile gives $(c - 24) / \sqrt{0.9} = 1.645$, and $c = 25.56$, which is greater than μ_0 . If $\bar{x} \leq 25.56$, the hypothesis is **accepted**. If $\bar{x} > 25.56$, it is rejected ■

Left-Sided Test The critical value c is obtained from the equation

$$P(\hat{X} \leq c)_{\mu=24} = \Phi\left(\frac{c-24}{\sqrt{0.9}}\right) = \alpha = 0.05.$$

Reverse engineering Tbl. ?? by looking for 0.95 percentile gives $c = 24 - \sqrt{0.9} \times 1.645 = 22.44$. If $\hat{x} \geq 22.44$, we accept the hypothesis. If $\hat{x} < 22.44$, we reject it ■

Two-Sided Test As the normal distribution is **symmetric**, we choose c_1 and c_2 equidistant from $\mu = 24$, say, $c_1 = 24 - k$ and $c_2 = 24 + k$, and determine k from:

$$P(24 - k \leq \hat{X} \leq 24 + k)_{\mu=24} = \Phi\left(\frac{k}{\sqrt{0.9}}\right) - \Phi\left(\frac{-k}{\sqrt{0.9}}\right) = 1 - \alpha = 0.95.$$

Looking for 0.975 in **Tbl. ??** gives $k/\sqrt{0.9} = 1.960$, therefore $k = 1.86$. This gives the values $c_1 = 24 - 1.86 = 22.14$ and $c_2 = 24 + 1.86 = 25.86$. If $\hat{x}$ is not smaller than c_1 and not greater than c_2 , we accept the hypothesis. Otherwise, we reject it ■

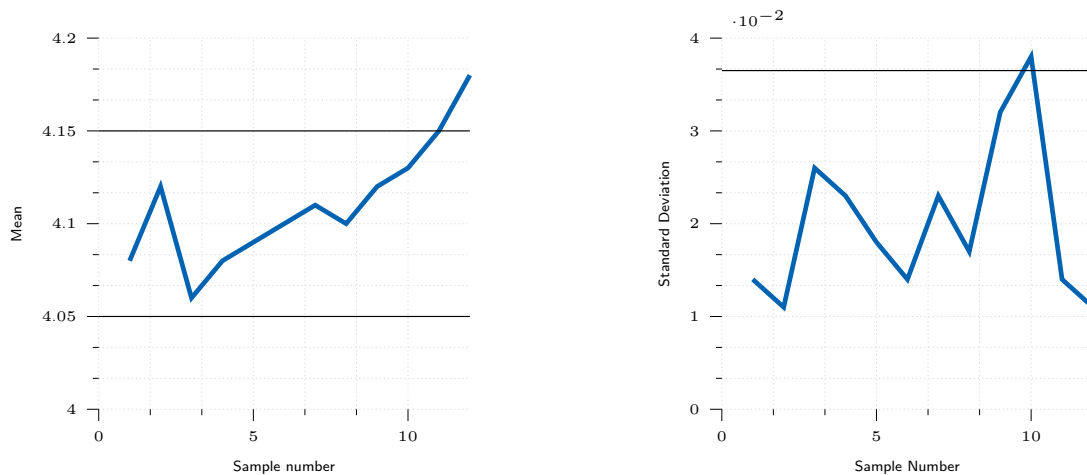


Figure 2.6: Control charts for the mean (upper part of figure) and the standard deviation in the case of the samples on p. 1089

2.5 Quality Control

The ideas on testing can be adapted and extended in various ways to serve basic practical needs in engineering and other fields. We will have a look at this in the remaining sections for some of the most important tasks solvable by statistical methods. As a first such area of problems, we will have a discussion on industrial quality control, a highly successful method used in various industries.

No production process is so perfect that all the products are completely alike. There is always a **small variation** caused by a great number of small, uncontrollable factors and must therefore be regarded as a chance variation. It is important to make sure the products have required values.³⁷ For this purpose one makes a test of the hypothesis that the points have the required property, say, $\mu = \mu_0$, where μ_0 is a required value. If this is done after an entire lot has been produced,³⁸ the test will tell us how good or how bad the products are, but it is obviously too late to alter undesirable results. It is much better to test during the production run. This is done at regular intervals of time³⁹ and is called quality control. Each time a sample of the same size is taken, in practice 3 to 10 times. If the hypothesis is rejected, we stop the production and look for the cause of the trouble.

If we stop the production process even though it is progressing properly, we make a **Type I error**. If we do not stop the process even though something is not in order, we make a **Type II error**. The result of each test is marked in graphical form on what is called a **control chart**.

This was proposed by W. A. Shewhart⁴⁰ in 1924 and makes quality control particularly effective.



⁴⁰Walter Andrew Shewhart (1891 - 1967) was an American physicist, engineer and statistician. He is sometimes also known as the grandfather of statistical quality control and also related to the Shewhart cycle.

³⁷for example, length, strength, or whatever property may be essential in a particular case

³⁸for example, a lot of 100,000 levels.

³⁹for example, every hour or half-hour

2.5.1 Control Chart for the Mean

An illustration and example of a control chart is given in the left part of **Fig. 2.6**. This control chart for the mean shows the **lower control limit** LCL, the **center control line** CL, and the **upper control limit** UCL. The two (2) control limits correspond to the critical values c_1 and c_2 . As soon as a sample mean falls outside the range between the control limits, we reject the hypothesis and assert that the production process is **out of control**. That is,

we assert that there has been a shift in process level and action is called for whenever a point exceeds the limits.

If we choose control limits that are too loose, we won't be able to detect process shifts. On the other hand, if we choose control limits that are too tight, we shall be unable to run the process because of frequent searches for nonexistent trouble. The usual significance level is $\alpha = 1\%$. Based on what we learned previously we see that in the case of the normal distribution the corresponding control limits (for $\alpha = 0.01$) for the mean are:

$$\text{LCL} = \mu_0 - 2.58 \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \text{UCL} = \mu_0 + 2.58 \frac{\sigma}{\sqrt{n}}.$$

Of course, it is worth mentioning here, σ is assumed to be **known**. If σ is unknown, we may compute the standard deviations of the first 20 or 30 samples and take their arithmetic mean as an approximation of σ .

Additional, more subtle controls are often used in industry. For instance, one observes the motions of the sample means above and below the centre-line, which should happen frequently. Accordingly, long runs⁴¹ of means all above, or all below the centring could indicate trouble.

⁴¹conventionally of length 7 or more

Sample Number	Sample Values					$\bar{x}$	s	R
1	4.06	4.08	4.08	4.08	4.10	4.080	0.014	0.04
2	4.10	4.10	4.12	4.12	4.12	4.112	0.011	0.02
3	4.06	4.06	4.08	4.10	4.12	4.084	0.026	0.06
4	4.06	4.08	4.08	4.10	4.12	4.088	0.023	0.06
5	4.08	4.10	4.12	4.12	4.12	4.108	0.018	0.04
6	4.08	4.10	4.10	4.10	4.12	4.100	0.014	0.04
7	4.06	4.08	4.08	4.10	4.12	4.088	0.023	0.06
8	4.08	4.08	4.10	4.10	4.12	4.096	0.017	0.04
9	4.06	4.08	4.10	4.12	4.14	4.100	0.032	0.08
10	4.06	4.08	4.10	4.12	4.16	4.104	0.038	0.10
11	4.12	4.14	4.14	4.14	4.16	4.140	0.014	0.04
12	4.14	4.14	4.16	4.16	4.16	4.152	0.011	0.02

Table 2.2: Twelve samples of five (5) values each (diameter of small cylinders, measured in mm)

2.5.2 Control Chart for the Variance

In addition to the mean, one often controls the variance, the standard deviation, or the range. To set up a control chart for the variance in the case of a normal distribution, we may employ the method we learned previously for determining control limits.

It is customary to use only one (1) control limit, namely, an upper control limit. Now we know the relation,

$$S^2 = \sigma_0^2 \frac{Y}{(n-1)},$$

where, because of our normality assumption, the random variable Y has a chi-square distribution with $n - 1$ degrees of freedom. Hence the desired control limit is

$$UCL = \left(\frac{\sigma^2 c}{n-1} \right) \quad (2.15)$$

where c is obtained from the equation

$$P(Y > c) = \alpha \quad \text{and} \quad P(Y \leq c) = 1 - \alpha$$

and the table of the chi-square distribution (given in Appendix) with $n - 1$ degrees of freedom. Here α (5% or 1%, say) is the probability that in a properly running process an observed value s^2 of S^2 is greater than the upper control limit.

If we wanted a control chart for the variance with both an upper control limit UCL and a lower control limit LCL, these limits would be

$$LCL = \frac{\sigma^2 c_1}{n-1} \quad \text{and} \quad UCL = \frac{\sigma^2 c_2}{n-1}. \quad (2.16)$$

where c_1 and c_2 are obtained from **Tbl. ??** to **Tbl. ??** with $n - 1$ d.f., and the equations

$$P(Y \leq c_1) = \frac{\alpha}{2} \quad \text{and} \quad P(Y \leq c_2) = 1 - \frac{\alpha}{2}.$$

2.5.3 Control Chart for the Standard Deviation

To set up a control chart for the standard deviation, we need an **upper control limit**.

$$UCL = \frac{\sigma \sqrt{c}}{\sqrt{n-1}}, \quad (2.17)$$

obtained from Eq. (2.15). For example, in **Table 2.2** we have $n = 5$. Assuming that the corresponding population is **normal** with standard deviation $\sigma = 0.02$ and choosing $\alpha = 1\%$, we obtain from the equation:

$$P(Y \leq c) = 1 - \alpha = 99\%$$

and with 4 d.f. the critical value $c = 13.28$ and from Eq. (2.17) the corresponding value:

$$UCL = \frac{0.02 \sqrt{13.28}}{\sqrt{4}} = 0.0365,$$

which is shown in the lower part of **Fig. 2.6**. A control chart for the standard deviation with both an upper and a lower control limit is obtained from Eq. (2.16).

2.5.4 Control Chart for the Range

Instead of the variance or standard deviation, one often controls the **range** R , where R is the largest sample value minus smallest sample value. It can be shown that in the case of the normal distribution, the standard deviation σ is proportional to the expectation of the random variable R^* for which R is an observed value, say, $\sigma = \lambda_n E(R^*)$ where the factor of proportionality λ_n depends on the sample size n and has the values: As R depends on two (2) sample values **only**, it

n	2	3	4	5	6	7	8	9	10
$\lambda_n E(R^*)$	0.89	0.59	0.49	0.43	0.40	0.37	0.35	0.34	0.32
n	12	14	16	18	20	30	40	50	
$\lambda_n E(R^*)$	0.31	0.29	0.28	0.28	0.27	0.25	0.23	0.22	

gives less information about a sample than s does. Clearly, the larger the sample size n is, the more information we lose in using R instead of s .

A practical rule is to use s when n is larger than 10.

Information: Application of Quality Control in Manufacturing

It is important to reiterate that quality control charts are not theoretical tools. Many industries today, from healthcare finance to manufacturing, use them to shape manufacturing practicalities. Let's explore their applications in two (2) critical areas:

- Mold design and,
- Mold manufacturing.

Mold Design

In manufacturing, accuracy is essential, especially when it comes to mold design. This is why, the slightest variation from the set specifications may result in delivering a substandard product. This is whereby control charts assist in monitoring the designs for consistency by tracking the essential dimensions and tolerance during the diverse stages of the design process.

During these crucial processes, various control charts can help engineers easily see whether a pattern is evident and determine if there is a problem with the measurement. For example, if the dimensions of a mold cavity lie outside the control limit more often, it will mean a need to alter something. Such an approach is helpful to avoid expensive mistakes and to ensure molds are built for proper part production.

Mold Manufacturing

Therefore, when it comes to the finalization of the mold design, it is time to move on to the next

level, which is manufacturing. Here, quality control charts are used effectively to keep the secured quality of the product along with a healthy flow of the process.

Through the regulation of parameters such as temperature, pressure, and cycle time, manufacturers are able to note any variance in the best conditions. For example, if the temperature of the molten material which is used in injection molding remains above the upper control limit, then it will cause some problems in the finished mold. This matter would be detected immediately by a control chart so that operators can change the temperature and avoid other difficulties.

There are so many benefits that can be obtained:

Reduce scrap and rework Control charts allow one to find out and correct the issues that result in the development of defective molds before more of them are developed and reduce on the amount of time and effort wasted.

Improve product quality This makes the production of molds to be more consistent since process monitoring is done frequently and continually hence fulfilling the customers' expectations.

Increase efficiency Through the control of these parameters, control charts facilitate the improvement of flow on processes by minimizing the cycle time. Enhance customer satisfaction: Quality molds that arrive on time and at a lower cost increase customer satisfaction, meaning more business.

2.6 Acceptance Sampling

Acceptance sampling is usually done when products leave the factory.⁴² The standard procedure in acceptance sampling is that a producer supplies to a consumer⁴³ a lot of N items.⁴⁴ The decision to accept or reject the lot is made by determining the number x of **defectives** in a sample of size n from the lot. The lot is accepted if $x \leq c$, where c is called the **acceptance number**, giving the allowable number of defectives. If $x > c$, the consumer rejects the lot. Clearly, producer and consumer must agree on a certain **sampling plan** giving n and c .

⁴²This is done in some cases even within the factory.

⁴³a buyer or wholesaler

⁴⁴a carton of screws, for instance

From the hyper-geometric distribution we see that the event A : “Accept the lot” has probability of:

$$P(A) = P(X \leq c) = \sum_{x=0}^c \binom{M}{x} \binom{N-M}{n-x} / \binom{N}{n} \quad (2.18)$$

where M is the number of defectives in a lot of N items. In terms of the **fraction defective** $\theta = M/N$ we can write Eq. (2.18) as:

$$P(A; \theta) = \sum_{x=0}^c \binom{N\theta}{x} \binom{N-N\theta}{n-x} / \binom{N}{n}. \quad (2.19)$$

$P(A; \theta)$ can assume $n+1$ values corresponding to $\theta = 0, 1/N, 2/N, \dots, N/N$; here, n and c are fixed. A monotone smooth curve through these points is called the **operating characteristic curve (OC curve)** of the sampling plan considered.

Information: Sampling Plan

Suppose that certain tool bits are packaged 20 to a box, and the following sampling plan is used.

A sample of two tool bits is drawn, and the corresponding box is accepted if and only if both bits in the sample are good.

In this case, $N = 20$, $n = 2$, $c = 0$, and Eq. (2.19) takes the form (a factor 2 drops out)

$$\begin{aligned} P(A; \theta) &= \binom{20\theta}{0} \binom{20-20\theta}{2} / \binom{20}{2} \\ &= \frac{(20-20\theta)(19-20\theta)}{380} \end{aligned}$$

The values of $P(A, \theta)$ for $\theta = 0, 1/20, 2/20, \dots, 20/20$ and the resulting OC curve are shown below.

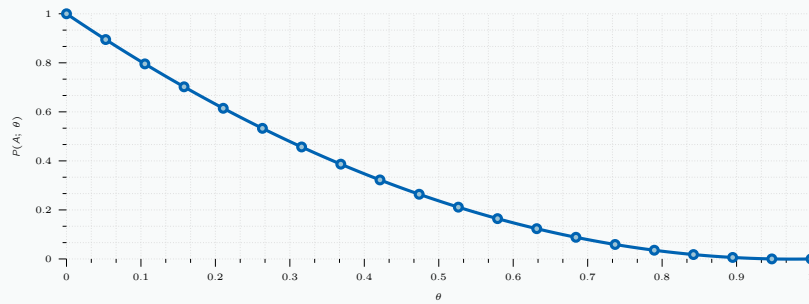


Figure 2.7: OC curve of the sampling plan $n = 2$ and $c = 0$ for lots of size $N = 20$.

In most practical cases θ will be small (less than 10%). Then if we take small samples compared to N , we can approximate Eq. (2.19) by the Poisson distribution; therefore

$$P(A; \theta) \sim e^{-\mu} \sum_{x=0}^c \frac{\mu^x}{x!} \quad \text{where} \quad \mu = n\theta. \quad (2.20)$$

Information: Sampling Plan using Poisson

Suppose that for large lots the following sampling plan is used. A sample of size $n = 20$ is taken. If it contains not more than one defective, the lot is accepted. If the sample contains two or more defectives, the lot is rejected. In this plan, we obtain from Eq. (2.20)

$$P(A; \theta) \sim e^{-20\theta} (1 + 20\theta) \quad \blacksquare$$

The corresponding OC curve is shown below

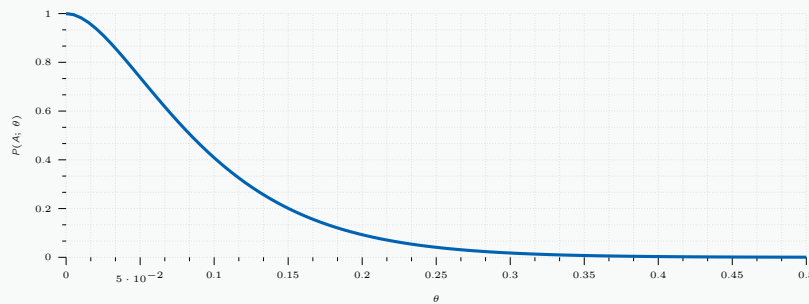


Figure 2.8: The OC curve with Poisson process applied to it.

2.6.1 Errors in Acceptance Sampling

We show how acceptance sampling fits into general test theory and what this means from a practical point of view. The producer wants the probability α of rejecting an acceptable lot⁴⁵ to be small. θ_0 is called the **acceptable quality level** (AQL). Similarly, the consumer wants the probability β of accepting an unacceptable lot⁴⁶ to be small.

⁴⁵a lot for which θ does not exceed a certain number θ_0 on which the two (2) parties agree

⁴⁶a lot for which θ is greater than or equal to some θ_1

In this scenario, θ_1 is called the **lot tolerance percent defective** or rejectable quality level (RQL). α is called **producer's risk** which would correspond to a Type I error discussed previously. β is called **consumer's risk** and corresponds to a Type II error. It can be shown that for large lots we can choose $\theta_0, \theta_1 (> \theta_0), \alpha, \beta$ and then determine n and c such that the OC curve runs very close to those recorded points. **Tbl. 2.3** shows the analogy between acceptance sampling and hypothesis testing.

Acceptance Sampling	Hypothesis Testing
Acceptable quality level(AQL) $\theta = \theta_0$	Hypothesis $\theta = \theta_0$
Lot tolerance percent defectives(LTPD) $\theta = \theta_1$	Alternative $\theta = \theta_1$
Allowable number of defectives c	Critical value c
Producer's risk α of rejecting a lot with $\theta \leq \theta_0$	Probability α of making a Type I error (significance level)
Consumer's risk β of accepting a lot with $\theta \geq \theta_1$	Probability β of making a Type II error

Table 2.3: Acceptance Sampling and Hypothesis Testing

2.6.2 Rectification

Rectification of a **rejected** lot means that the lot is inspected item by item and all defectives are removed and replaced by non-defective items.⁴⁷ If a production turns out $100\theta\%$ defectives, then in K lots of size N each, $KN\theta$ of the KN items are **defectives**.

⁴⁷This may be too expensive if the lot is cheap. If this were the case, the lot may be sold at a cut-rate price or scrapped.

Now $KP(A; \theta)$ of these lots are accepted. These contain $KPN\theta$ defectives, whereas the rejected and rectified lots contain **no** defectives, because of the rectification. Hence after the rectification the fraction defective in all K lots equals $KPN\theta/KN$. This is called the **average outgoing quality** (AQO), which is expressed as:

$$AOQ(\theta) = \theta P(A; \theta).$$

Fig. 2.9 shows an example. Since $AOQ(0) = 0$ and $P(A; 1) = 0$, the AOQ curve has a maximum at some $\theta = \theta^*$, giving the **average outgoing quality limit** (AOQL). This is the worst average quality that may be expected to be accepted under rectification.

Information: Application of Acceptance Sampling in Industry

The applications are diverse and span numerous industries, each with its unique quality control requirements and challenges.

Manufacturing Widely used for evaluating the quality of raw materials, components, and finished products. Examples include inspecting batches of electronic components, assessing the dimensional accuracy of machined parts, or verifying the structural integrity of construction materials.

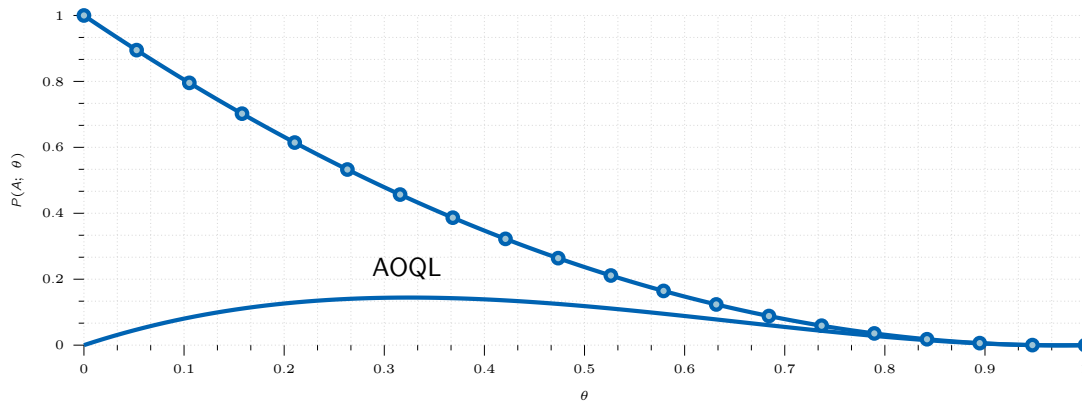


Figure 2.9: OC curve and AOQL curve for the sampling plan.

Electronics The electronics industry relies heavily on it to ensure the quality and reliability of components, circuits, and devices. Common applications include inspecting printed circuit boards for defects, testing the performance of semiconductors, and evaluating the functionality of electronic assemblies.

Pharmaceuticals Plays a critical role in ensuring the safety and efficacy of drugs and medical devices. Companies employ acceptance sampling techniques to inspect batches of raw materials, active ingredients, and finished pharmaceutical products, verifying compliance with stringent quality standards and regulatory requirements.

2.7 Goodness of Fit

To test for goodness of fit⁴⁸ means that we wish to test that a certain function $F(x)$ is the distribution function of a distribution from which we have a sample $x_1, \dots, x_n$. Then we test whether the **sample distribution function** $\tilde{F}(x)$ defined as:

$$\tilde{F}(x) = \text{Sum of the relative frequencies of all sample values } x_j \text{ not exceeding } x,$$

fits $\tilde{F}(x)$ **sufficiently well**. If this is so, we shall **accept** the hypothesis that $\tilde{F}(x)$ is the distribution function of the population; else, we shall **reject the hypothesis**.

This test is of considerable practical importance, and it differs in character from the tests for parameters (μ , σ^2 , etc.) considered thus far.

To test in that fashion, we have to know how much $\tilde{F}(x)$ can differ from $F(x)$ if the hypothesis is **true**. Hence we must first introduce a quantity which measures the deviation of $\tilde{F}(x)$ from $F(x)$, and we must know the probability distribution of this quantity under the assumption that the hypothesis is true.

Then we proceed as follows.

We determine a number, let's use c , such that, if the hypothesis is **true**, a deviation greater than c has a small preassigned probability. If, nevertheless, a deviation greater than c occurs, we have reason to doubt that the hypothesis is true and we reject it. On the other hand, if the deviation does **NOT** exceed c , so that $\tilde{F}(x)$ approximates $F(x)$ sufficiently well, we accept the hypothesis.

Of course, if we accept the hypothesis, this means that we have insufficient evidence to reject it, and this does not exclude the possibility that there are other functions that would not be rejected in the test.

In this respect the situation is quite similar to hypothesis testing we talked previously.

The following text-block shows a test of that type, which was introduced by *R. A. Fisher*. This test is justified by the fact that if the hypothesis is true, then χ_0^2 is an observed value of a random variable whose distribution function approaches that of the chi-square distribution with $K - 1$ degrees of freedom⁴⁹ as n approaches infinity. The requirement that at least five (5) sample values lie in each interval results from the fact that for finite n that random variable has only approximately a chi-square distribution.

If the sample is so small that the requirement cannot be satisfied, one may continue with the test, but then use the result **with caution**.

⁴⁸In literature, this method also means χ^2 -Test.

Historical Anecdote

During the 19th century, statistical analytical methods were mainly applied to biological data and it was customary for researchers to assume observations followed a **normal distribution**, such as Sir George Airy and Mansfield Merriman, whose works were criticized by Karl Pearson in his 1900 paper.

At the end of the 19th century, Pearson noticed the existence of significant skewness within some biological observations. To model the observations regardless of being normal or skewed, Pearson, in a series of articles published from 1893 to 1916, devised the Pearson distribution, a family of continuous probability distributions, which includes the normal distribution and many skewed distributions, and proposed a method of statistical analysis consisting of using the Pearson distribution to model the observation and performing a test of goodness of fit to determine how well the model really fits to the observations.

⁴⁹or $K - r - 1$ degrees of freedom if r parameters are estimated.

Theory 2.22: Chi-square Test for $F(x)$ being the Distribution Function of a Population

1. Subdivide the x -axis into n intervals $x_1, \dots, x_n$ such that each interval contains at least five (5) values of the given sample $x_1, \dots, x_n$.

Determine the number b_j of sample values in the interval I_j , where $j = 1, \dots, K$. If a sample value lies at a common boundary point of two (2) intervals, add 0.5 to each of the two (2) corresponding b_j .

2. Using $F(x)$, calculate the probability p_j that the random variable X under consideration assumes any value in the interval I_j , where $j = 1, \dots, K$. Then, calculate

$$e_j = np_j.$$

This is the number of sample values **theoretically expected** in I_j if the hypothesis is true.

3. Compute the deviation:

$$\chi_0^2 = \sum_{j=1}^K \frac{(b_j - e_j)^2}{e_j}.$$

4. Choose a significance level such as 5%, 1%, or the like.
5. Determine the solution c of the equation

$$P(\chi^2 \leq c) = 1 - \alpha.$$

from the table of the chi-square distribution with $K - 1$ degrees of freedom **Tbl. ??** to **Tbl. ??**.

If r parameters of $F(x)$ are unknown and their maximum likelihood estimates are used, then use $K - r - 1$ degrees of freedom, instead of $K - 1$.

If $\chi_0^2 \leq c$, accept the hypothesis. If $\chi_0^2 > c$, reject the hypothesis.

Exercise 2.10 Printed Circuit Boards

The number of defects in printed circuit board is hypothesized to follow a Poisson distribution. A random sample of $n = 60$ printed boards have been collected, and following number of defects were observed

Number of Defects	Observed Frequency
0	32
1	15
2	9
3	4

SOLUTION The mean of the assumed Poisson distribution in this example is unknown and must be estimated from the sample data. The estimate of the mean number of defects per board is the sample average, that is:

$$(32 \times 0 + 15 \times 1 + 9 \times 2 + 4 \times 3) / 60 = 0.75$$

From the Poisson distribution with parameter 0.75, we may compute p_i , the theoretical, hypothesized probability associated with the i^{th} class interval. Since each class interval corresponds to a particular number of defects, we may find the p_i as follows:

$$\begin{aligned} p_1 &= P(X = 0) = \frac{e^{-0.75}(0.75)^0}{0!} = 0.472 \\ p_2 &= P(X = 1) = \frac{e^{-0.75}(0.75)^1}{1!} = 0.354 \\ p_3 &= P(X = 2) = \frac{e^{-0.75}(0.75)^2}{2!} = 0.133 \\ p_4 &= P(X \geq 3) = 1 - (p_1 + p_2 + p_3) = 0.041 \end{aligned}$$

The expected frequencies are computed by multiplying the sample size $n = 60$ times the probabilities p_i . That is, $e_i = np_i$. The expected frequencies follow:

Number of Defects	Probability	Expected Frequency
0	0.472	28.32
1	0.354	21.24
2	0.133	7.98
3 (or more)	0.041	2.46

Since the expected frequency in the last cell is less than 3, we combine the last two cells:

NOTE: Categories with expected frequency is combined because the Chi-square test would not work if the frequency is less than 5. If the sample size is too small the chi-square value is over-estimated and if it is too large chi-square value is under-estimated. Hence why we combine with the category with the lowest frequency.

Since the expected frequency in the last cell is less than 3, we combine the last two cells:

Number of Defects	Probability	Expected Frequency
0	0.32	28.32
1	0.15	21.24
2 (or more)	0.13	10.44

Now, the chi-square test will have $k - p - 1 = 3 - 1 - 1 = 1$ degree of freedom, because the mean of the Poisson distribution was estimated from the data.

The hypothesis-testing procedure may now be applied using $\alpha = 0.05$,

1. The variable of interest is the form of the distribution of defects in printed circuit boards.
2. θ_0 The form of the distribution of defects is Poisson.
3. θ_1 The form of the distribution of defects is not Poisson.
4. Test statistic is:

$$\chi_0^2 = \sum_{j=1}^k \frac{(b_j - e_j)^2}{e_j}$$

5. Reject H_0 if $\chi_0^2 > \chi_{0.05,1}^2 = 3.84$.

6. Time to calculate χ_0^2 :

$$\chi_0^2 = \frac{(32 - 28.32)^2}{28.32} + \frac{(15 - 21.24)^2}{21.24} + \frac{(13 - 10.44)^2}{10.44} = 2.94$$

7. As $\chi_0^2 = 2.94 < \chi_{0.05,1}^2 = 3.84$, we are unable to reject the null hypothesis that the distribution of defects in printed circuit boards is Poisson.

1. Leonbloy (<https://math.stackexchange.com/users/312/leonbloy>). *Maximum Likelihood Estimate and Second derivative test?* Mathematics Stack Exchange. URL:<https://math.stackexchange.com/q/2241123> (version: 2017-04-19). eprint: <https://math.stackexchange.com/q/2241123>.
2. John, R. Maximum likelihood estimation of discrete control processes. *SIAM journal on control and optimization* **26**, 1006–1024 (1988).
3. Aït-Sahalia, Y. & Kimmel, R. Maximum likelihood estimation of stochastic volatility models. *Journal of financial economics* **83**, 413–452 (2007).
4. Teka, K. H. Parameter estimation of the Black-Scholes-Merton model (2013).
5. Stewart, W. C. & Thompson, E. A. Improving estimates of genetic maps: a maximum likelihood approach. *Biometrics* **62**, 728–734 (2006).
6. Chakravarti, A., Lasher, L. K. & Reefer, J. E. A maximum likelihood method for estimating genome length using genetic linkage data. *Genetics* **128**, 175–182 (1991).
7. Neyman, J. *On the two different aspects of the representative method: the method of stratified sampling and the method of purposive selection in Breakthroughs in statistics: Methodology and distribution* 123–150 (Springer, 1992).
8. Neyman, J. Outline of a theory of statistical estimation based on the classical theory of probability. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* **236**, 333–380 (1937).
9. Neyman, J. & Pearson, E. S. IX. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character* **231**, 289–337 (1933).

Part II

Introduction to Machine Learning

Acronyms

		L
LHS	Left Hand Side	
		R
RHS	Right Hand Side	

